EE5360 | Practical Challenges in Image Analysis

[MUD] MIC Determination Using CV

Pre-final Presentation

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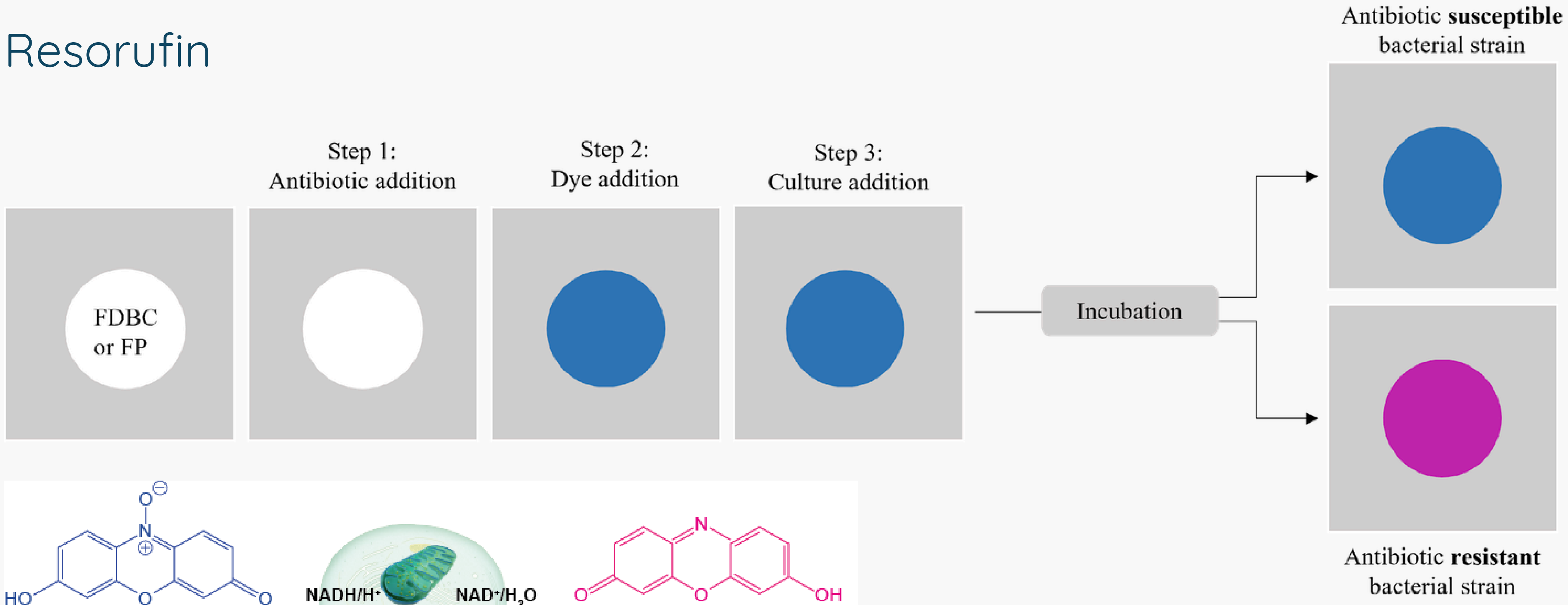
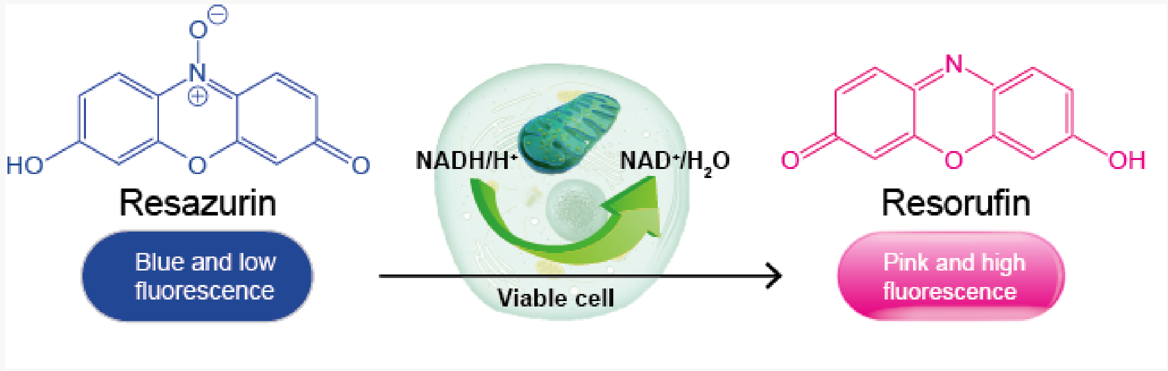
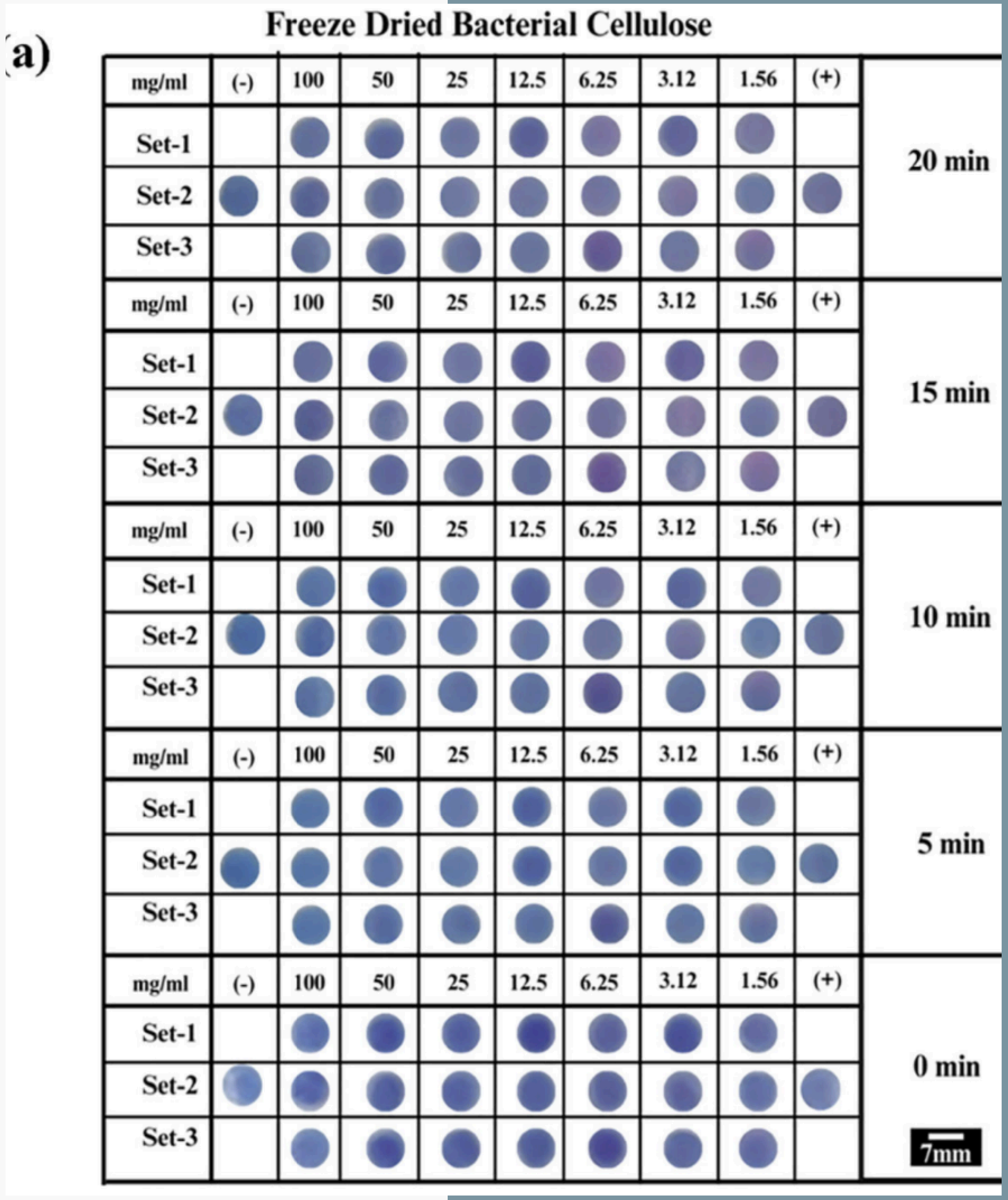
Recap

Current scenario:

Observing a redox reaction where living cells convert blue/purple Resazurin into pink Resorufin







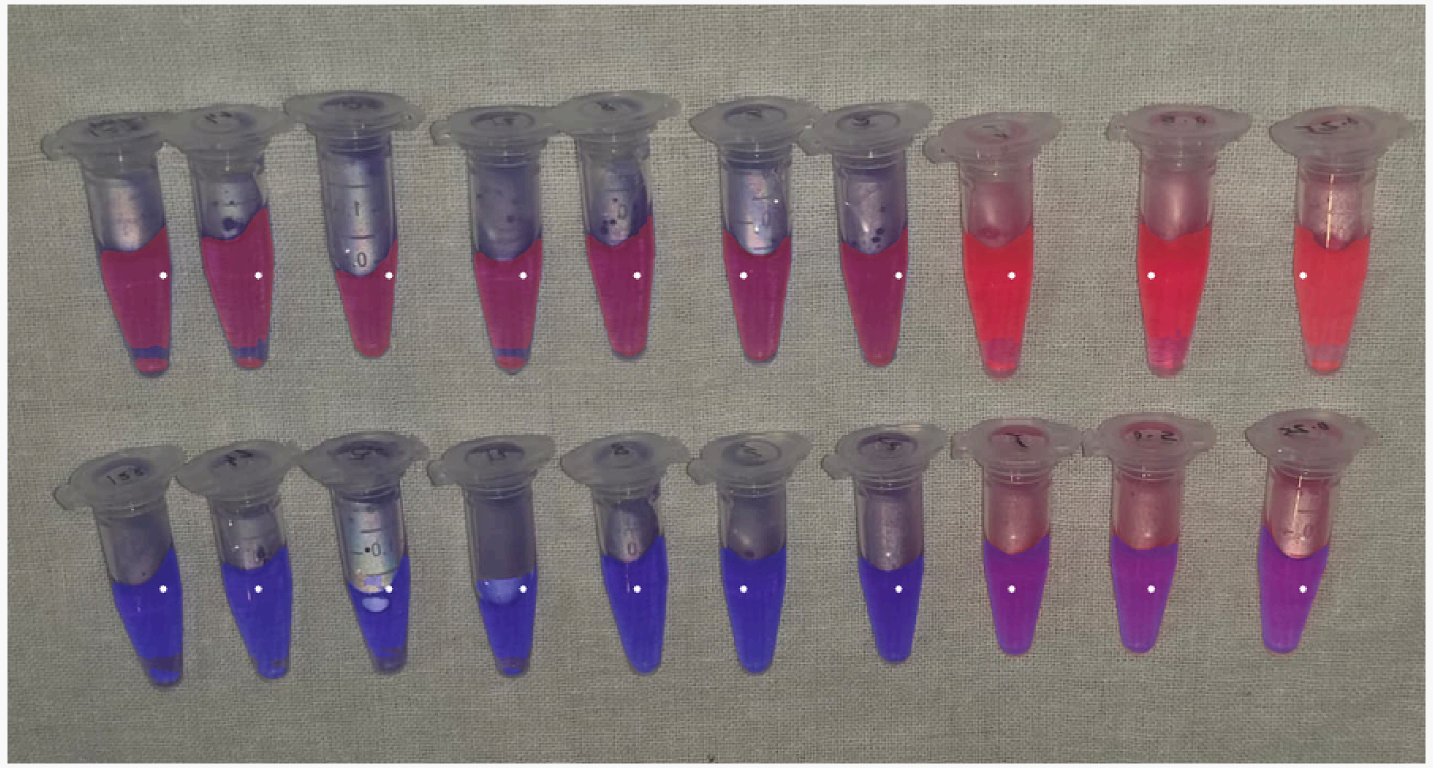
Literature Survey & Related Work

- Freeze-Dried Cellulose for AST (2026)
  - Source: Sarita C, Khandelwal M, Duraiswamy S. Freeze-Dried Bacterial Cellulose-Based Point-of-Care Device for Antibiotic Susceptibility Testing (AST) (2026).
  - Relevance: context, background, need.
- Smartphone-Based Concentration Classification
  - Source: ACS Omega (2021).
  - Relevance: Establishes that smartphone images can be used for "coarse" classification of chemical concentrations. This validates our approach of using computer vision on simple camera images rather than expensive spectrophotometers.
- ML-Assisted Chemiluminescence
  - Source: Talanta (2023).
  - Relevance: Highlights the power of Machine Learning (ML) in discriminating subtle signal changes (chemiluminescence) that human eyes might miss. We apply this same logic to subtle color shifts (Purple to Pink).

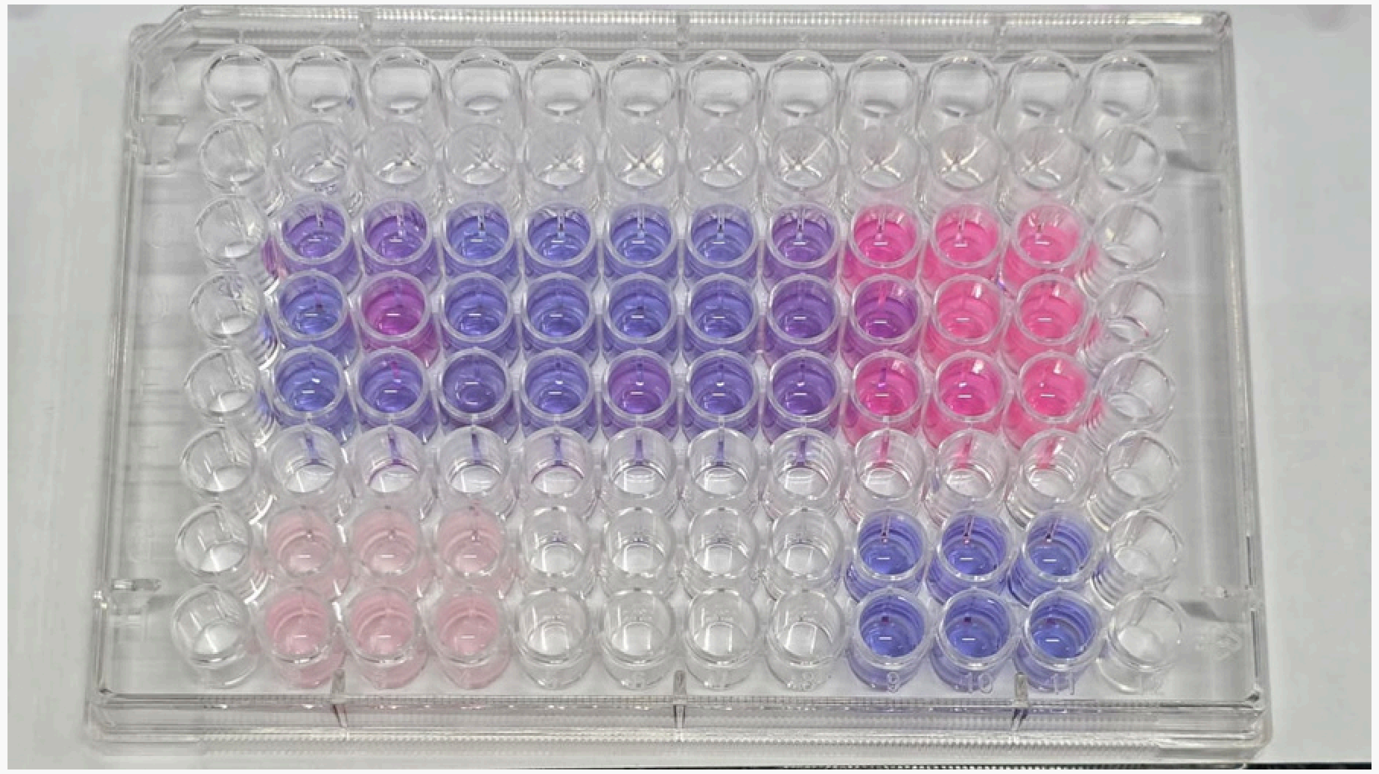
Problem Statement

- The Cost Barrier in Current AST: Standard Antibiotic Susceptibility Testing (AST) relies on Optical Density (OD) and absorbance values, requiring expensive microplate readers (costing upwards of ₹10 Lakhs). This heavily limits Point-of-Care (PoC) accessibility in resource-limited settings.
- The Colorimetric Solution: Using a Resazurin indicator enables visual detection (Purple to Pink). This is more intuitive for clinicians and essential for future Freeze-Dried Bacterial Cellulose (FDBC) Point-of-Care devices.
- The Intermediate Challenge: Standardizing computer vision models on well-plates and vials first to ensure accuracy before transitioning to the more complex FDBC substrate.
- Stepwise Project Goals:
    - Quality Control: Automatically detect anomalous data or "bad runs" (e.g., contamination or technical failure).
    - Binary Classification: Distinguish between "Live/Growth" and "Dead/Inhibition" states in any given sample.
    - MIC Determination: Combine these findings to identify the absolute Minimum Inhibitory Concentration across a dilution series.

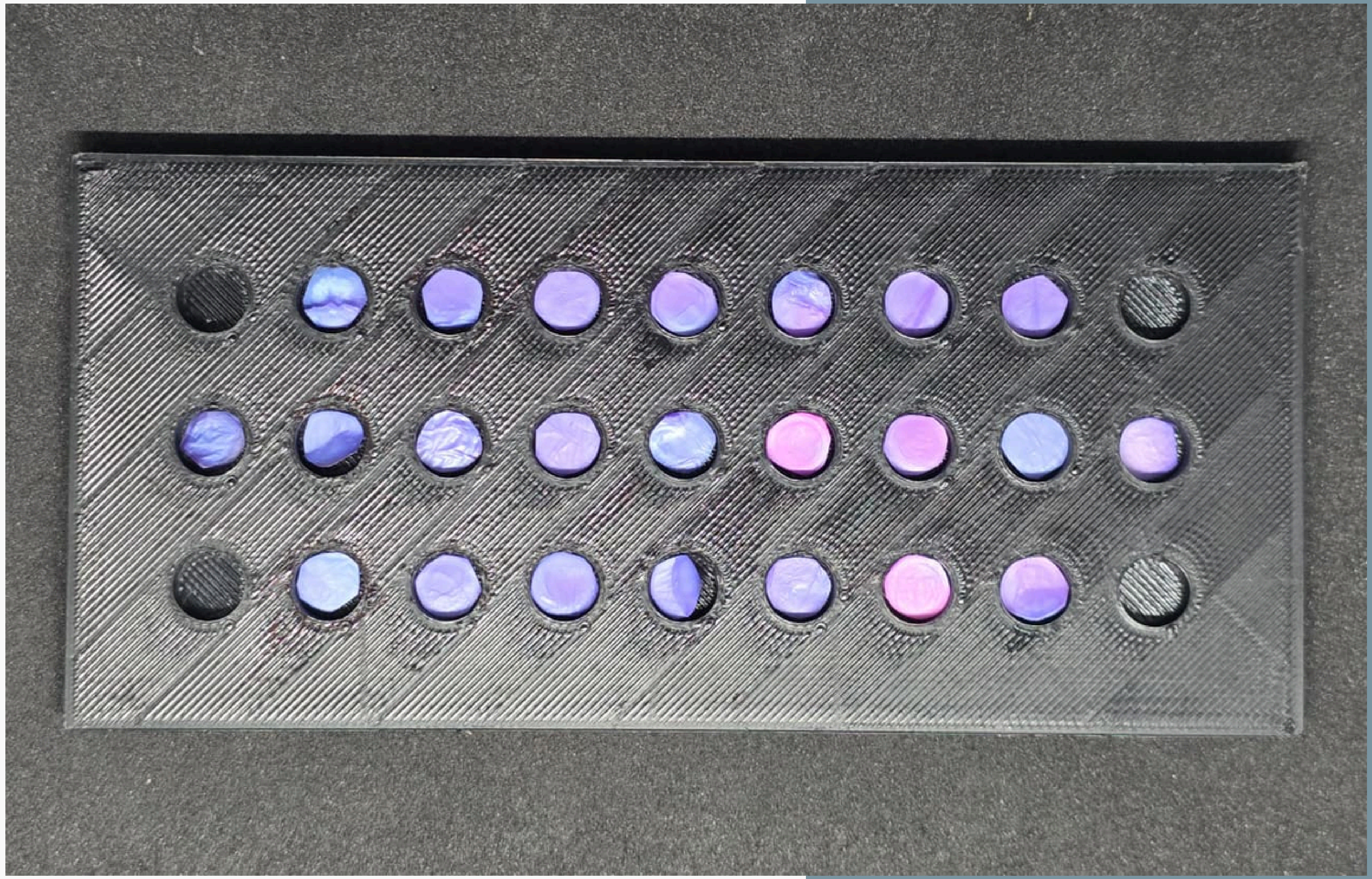
Examples of Images Provided



Vials

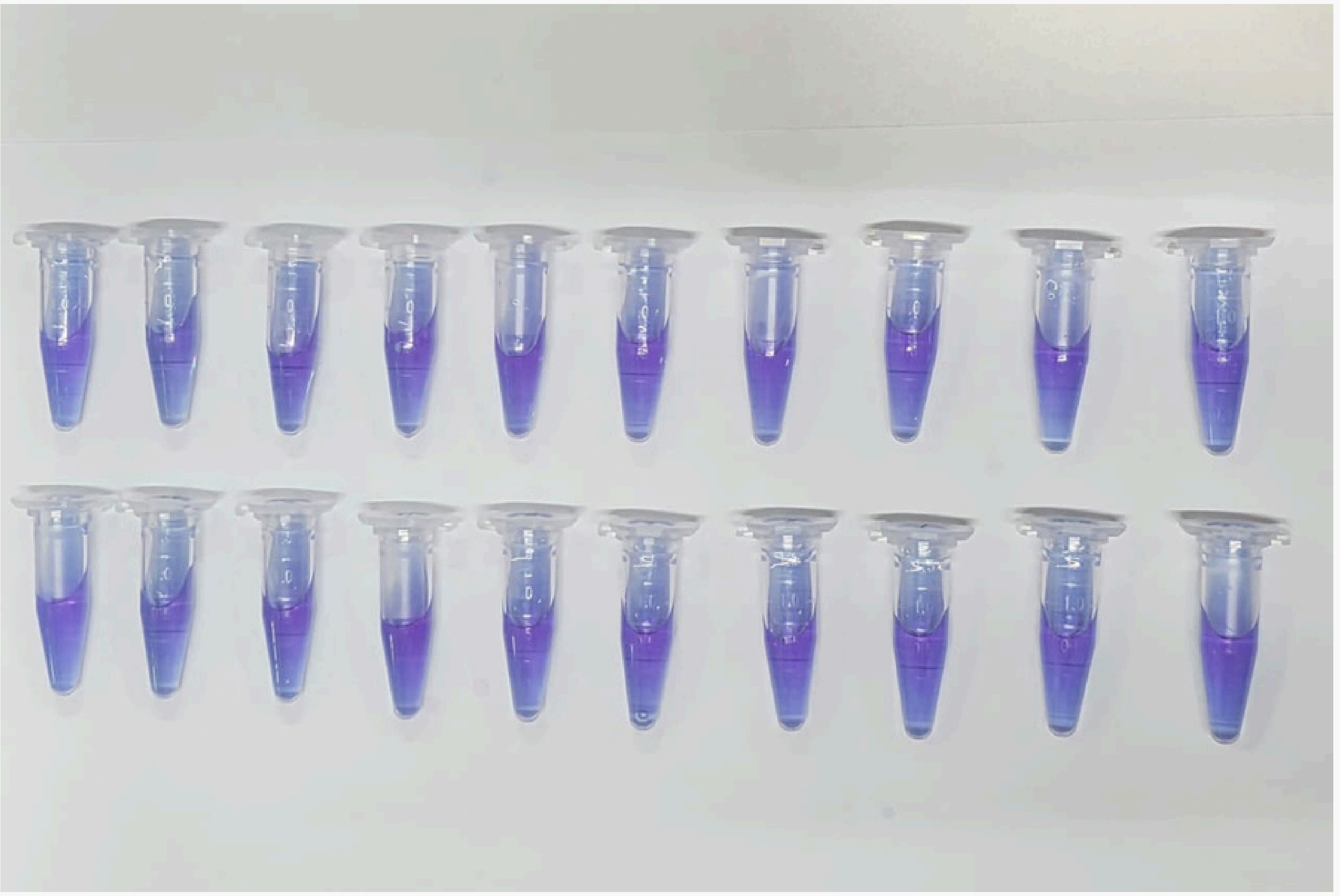


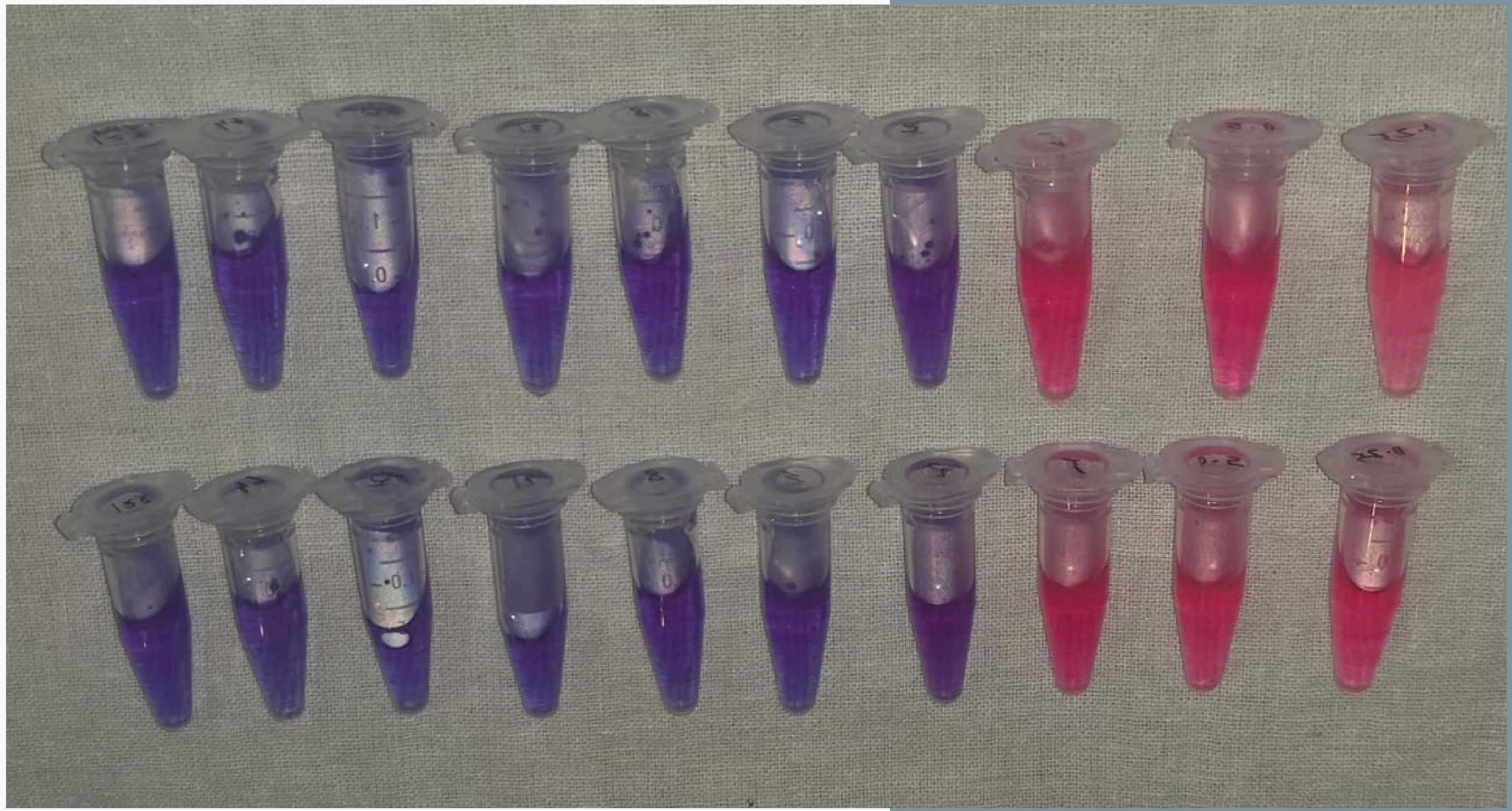
Well-plates



FDBC

Vials

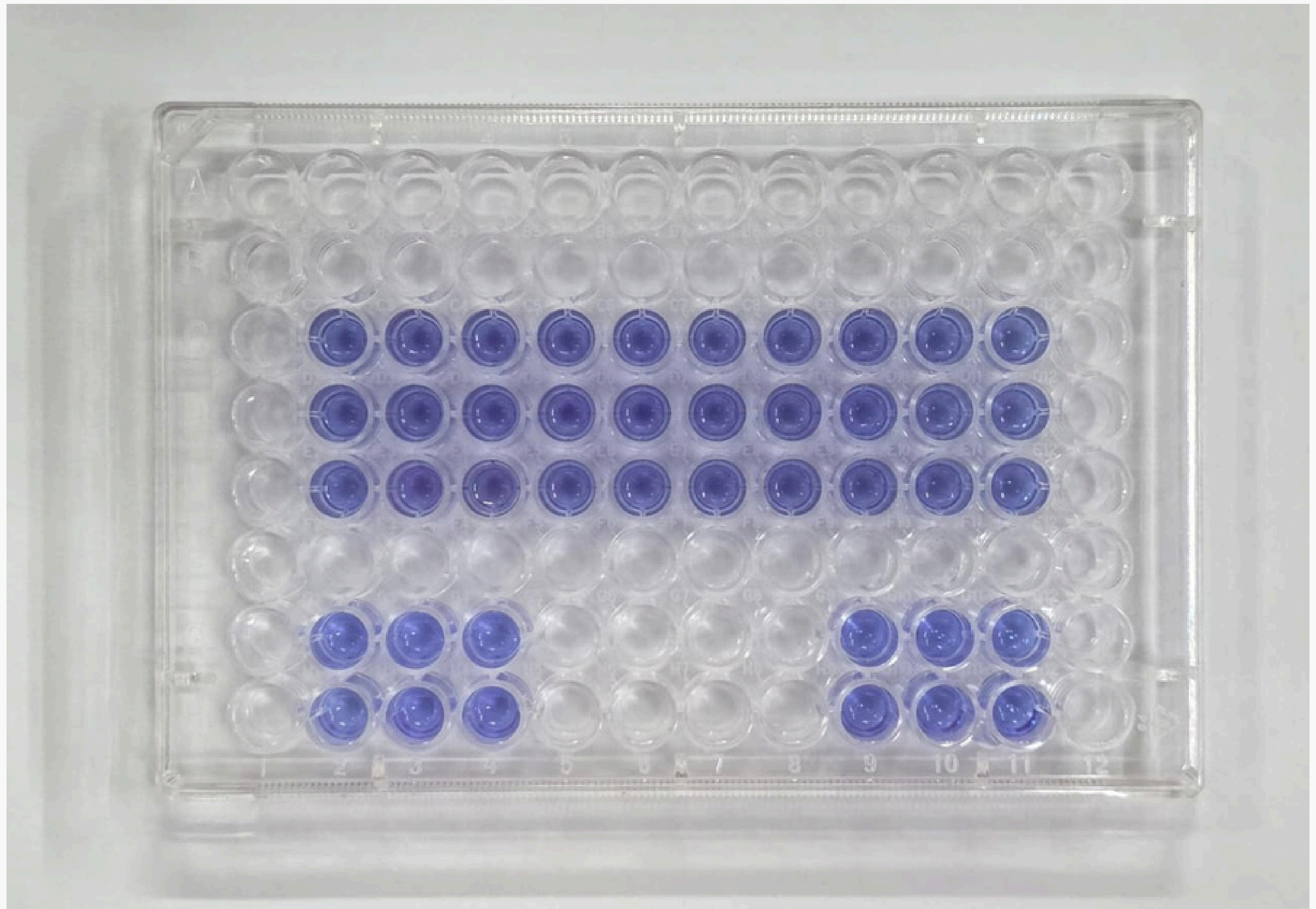




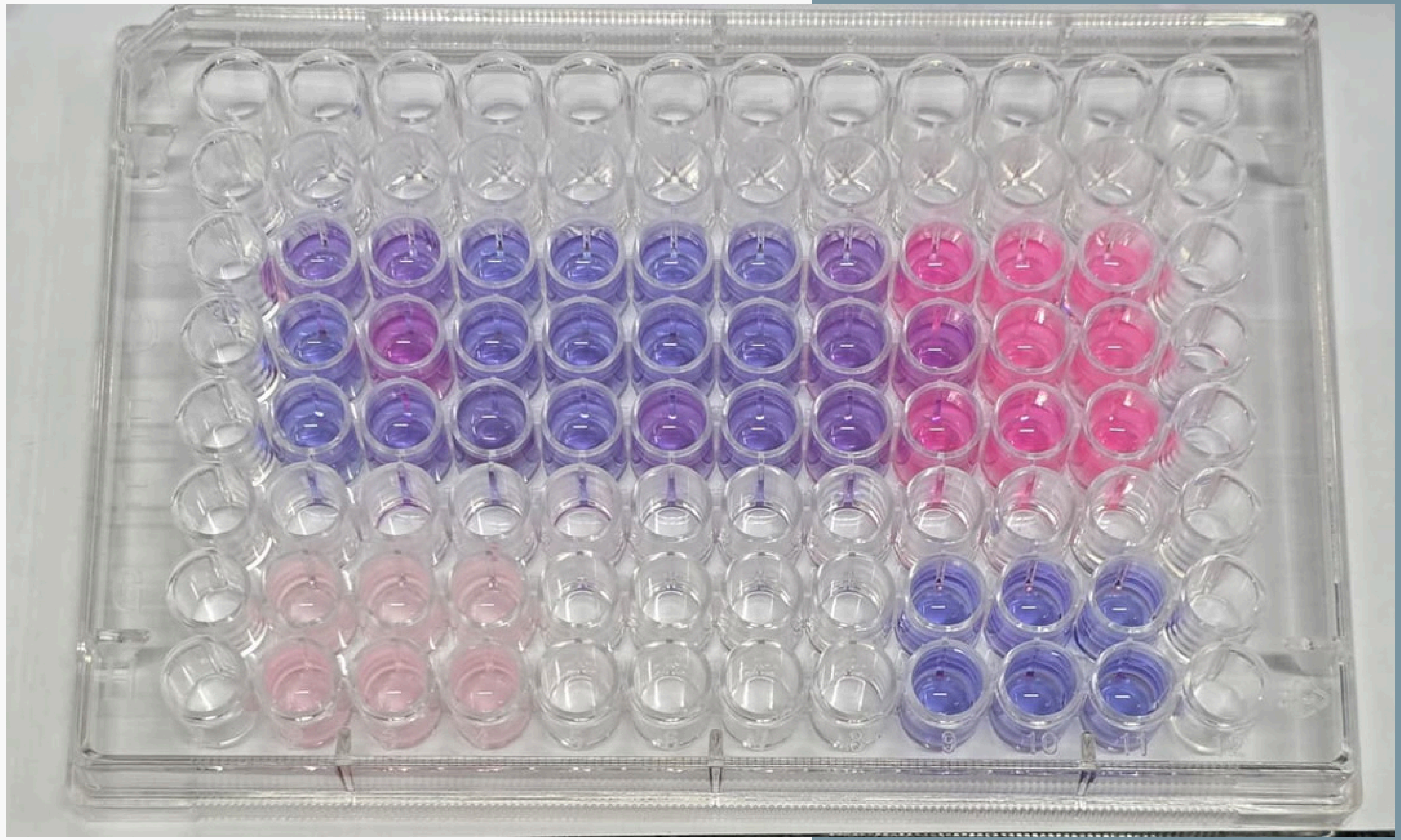
0th min (over 4 hours)

75th min (over 4 hours)

Well-plates

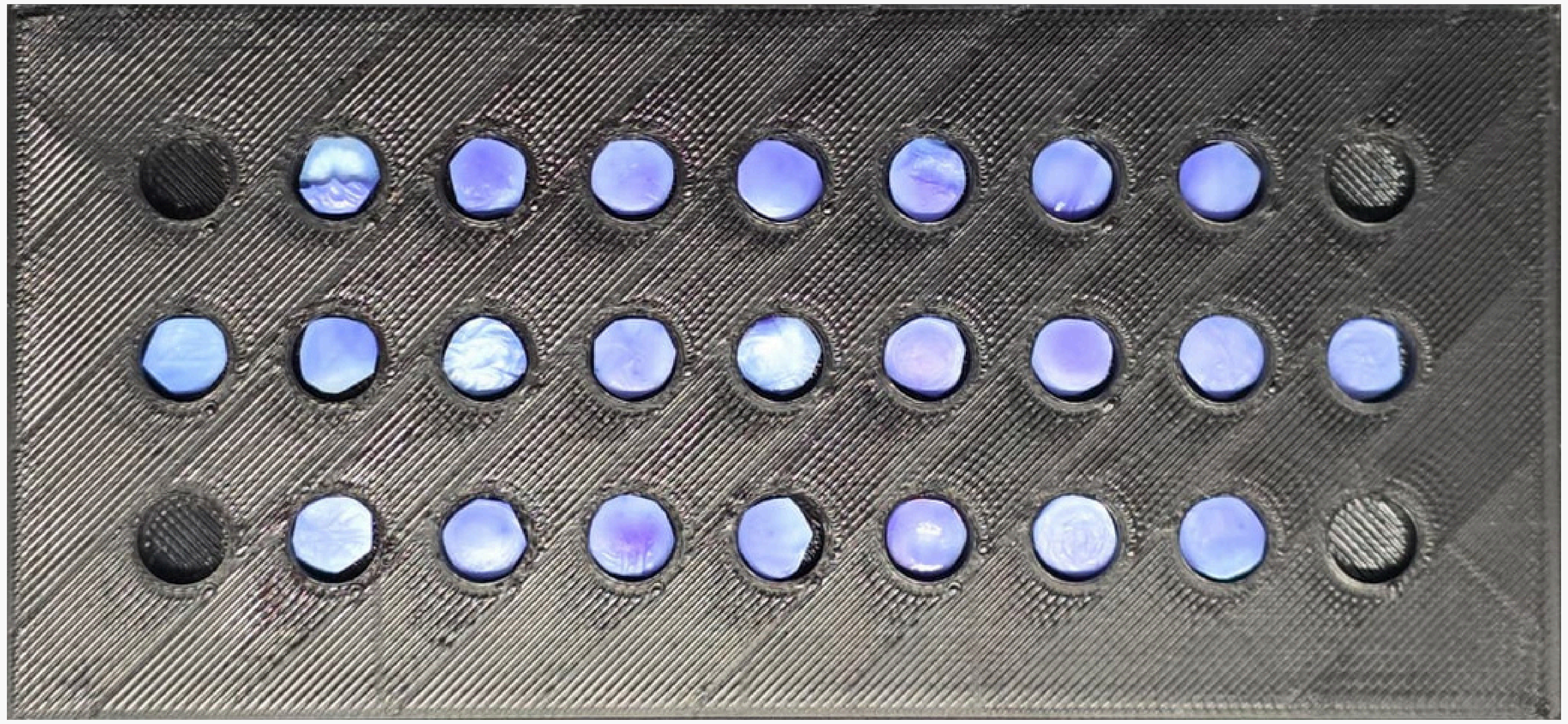


0th min (over 4 hours)

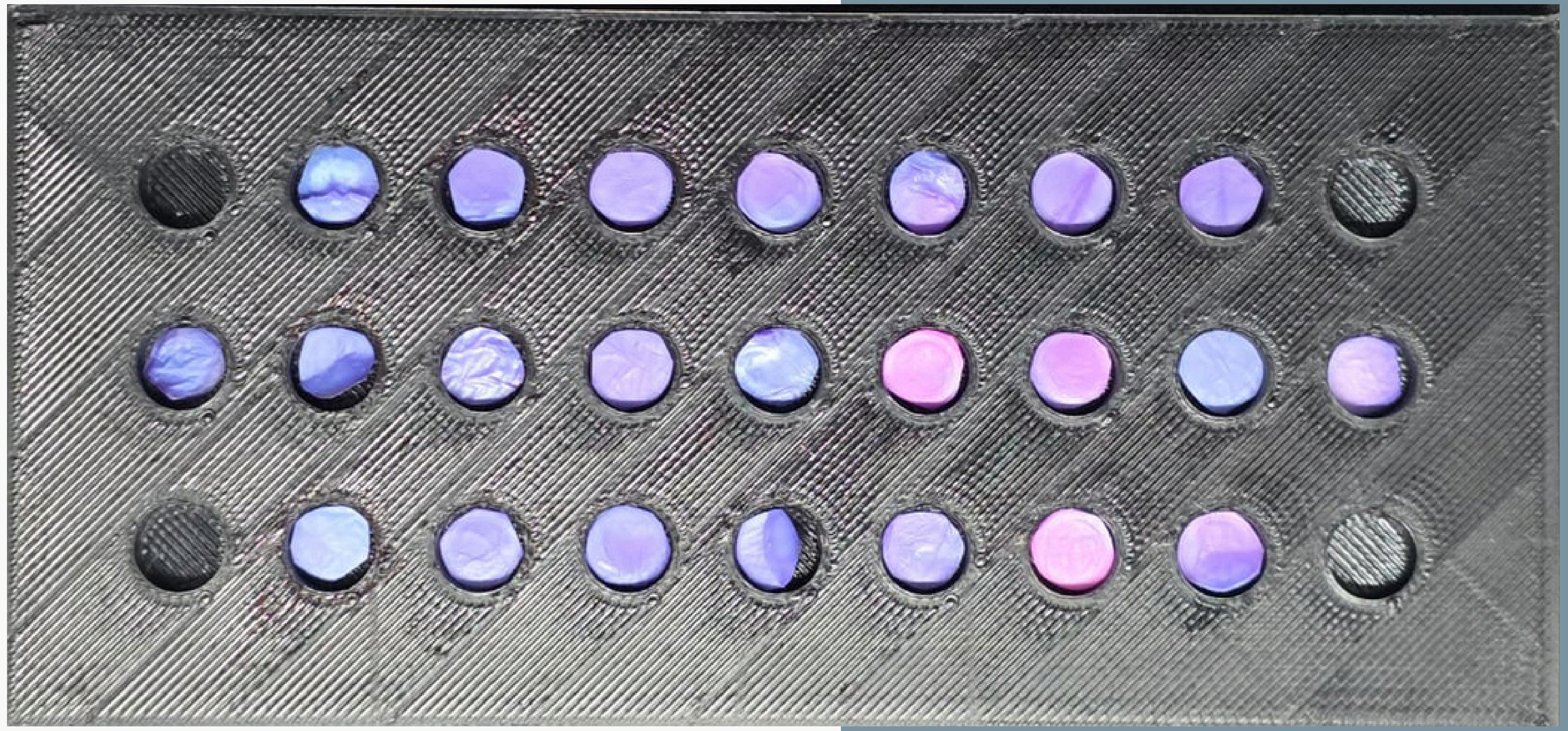


70th min (over 4 hours)

Freeze-dried Bacterial Cellulose



0th min



25th min

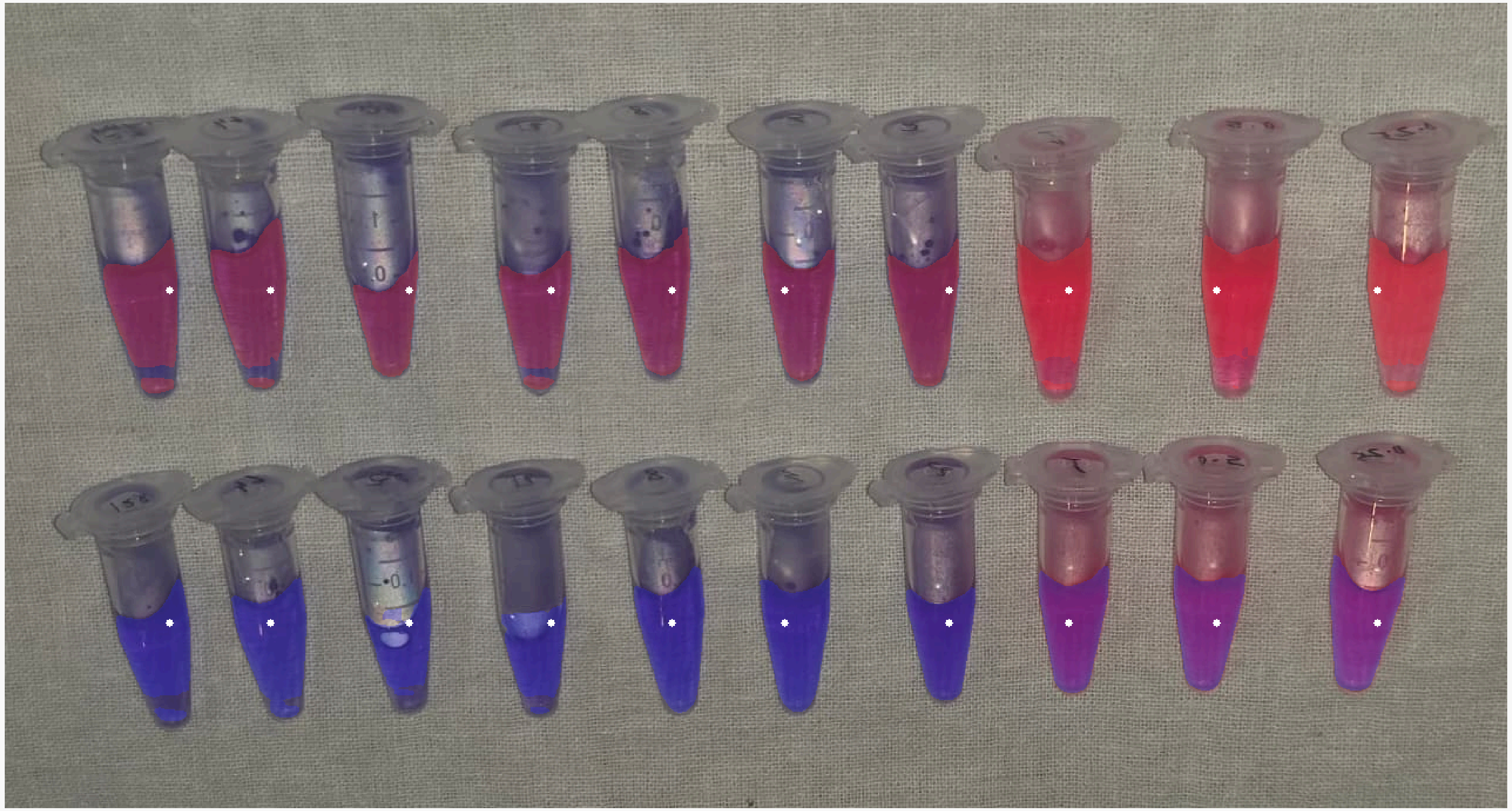
Quantifying Color Change

- We need to track the ‘pinkness’ of the experiment contents.
- The a* channel: We convert images to the CIELAB color space and specifically analyze the a* channel, which provides a quantitative measure of "Pinkness" independent of lighting brightness.
- Data Aggregation: For each segmented region, we calculate the median a* value to represent the growth state of that specific antibiotic concentration.

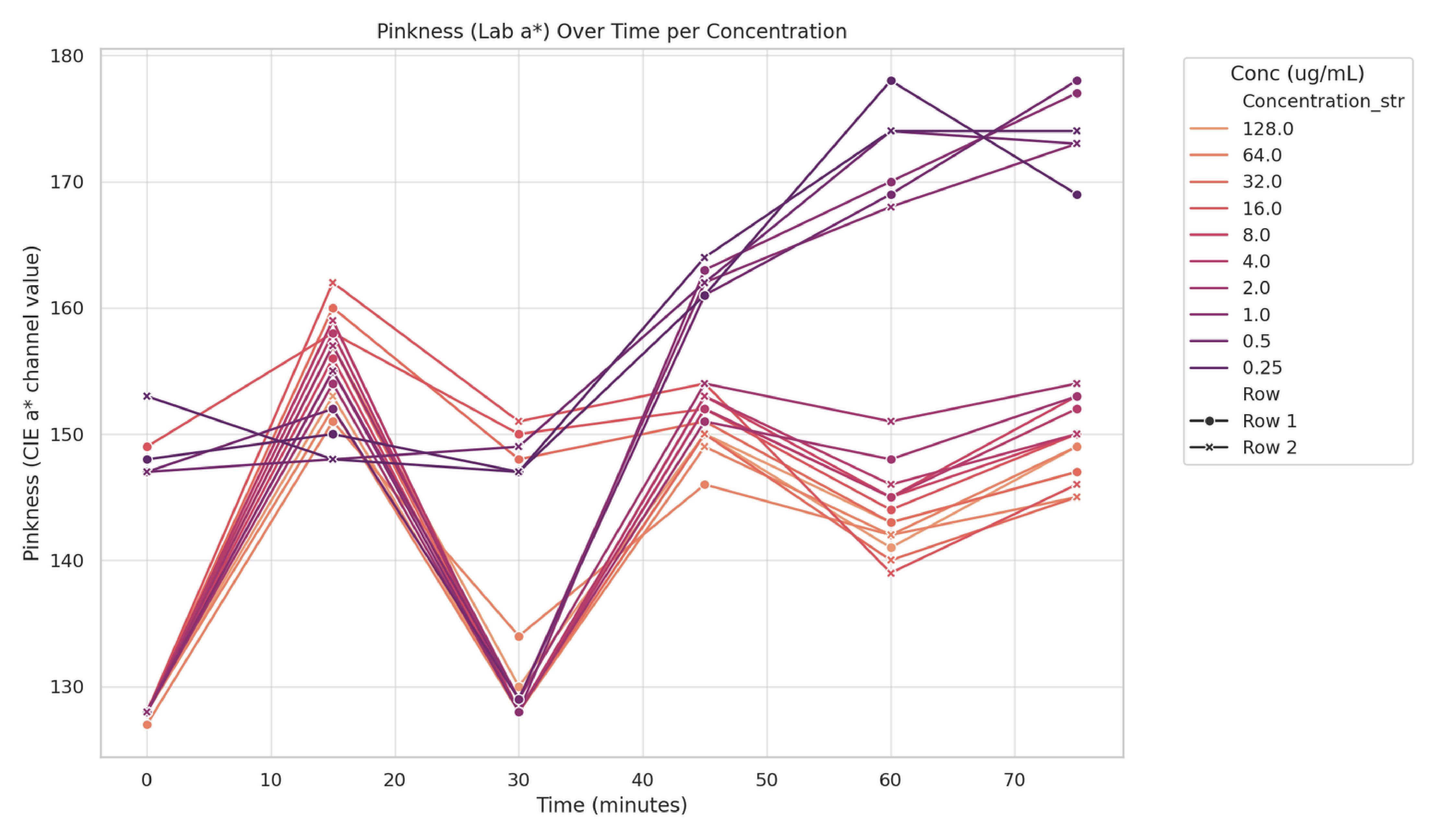
Segmentation with "Signal-Guided ML"

- The Challenge: Transparent test tubes and fluid reflections create ambiguous boundaries that confuse standard edge detection techniques and segmentation models.
- Signal Extraction: We use the Saturation channel (HSV) to isolate the experiment contents - the colored fluid (pink/purple) stands out with high saturation, while the white background and gray shadows are naturally filtered out.
- ML-Powered Precision: We use SAM, but guide it with the robust saturation signals (centers and rough bboxes of the fluid regions) and ensure precise masking across all 20 tubes.

Results...



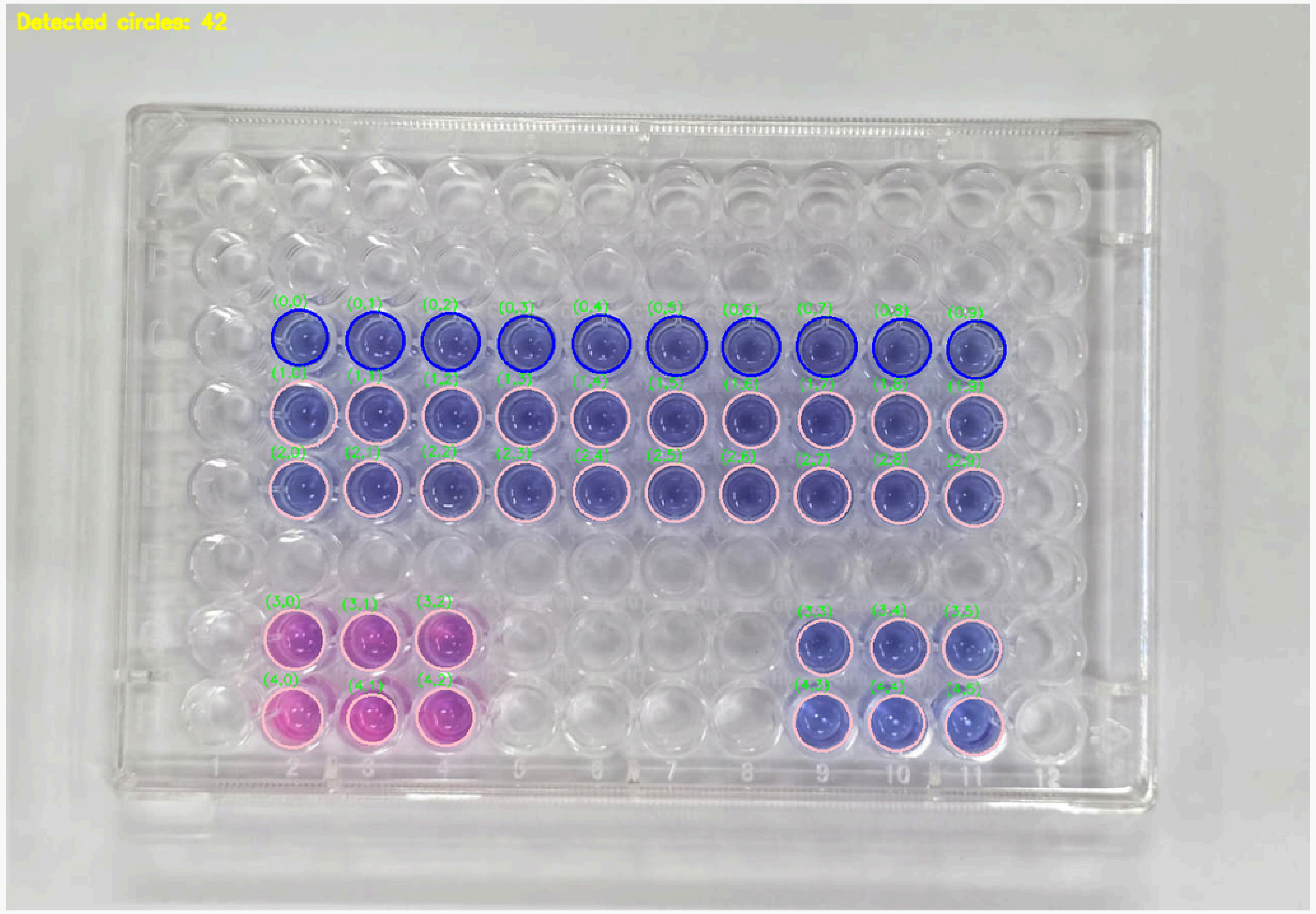
Results…

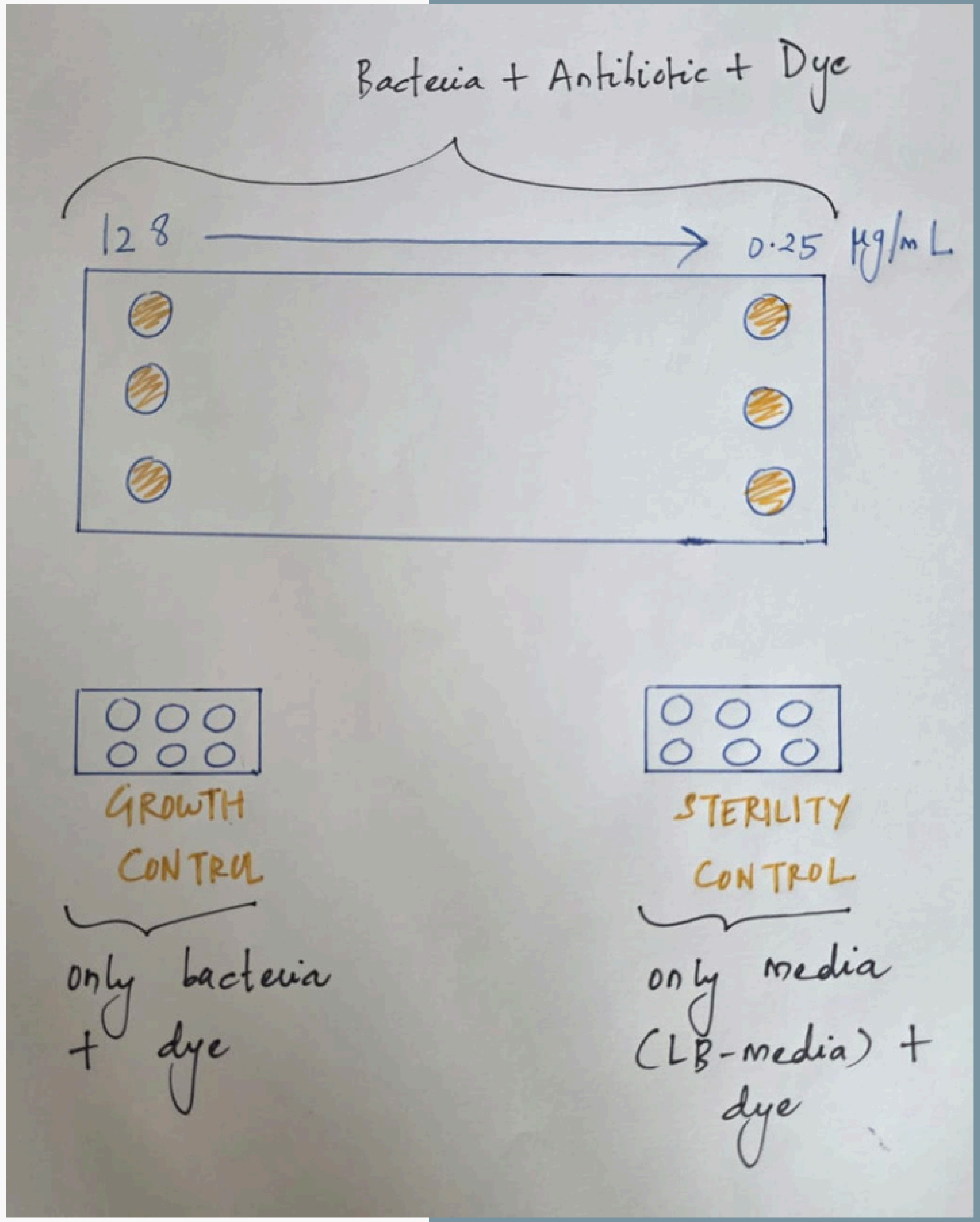


Pinkness (Lab a*) Over Time per Concentration
Pinkness (CIE a* channel value)
180
170
160
150
140
130
0
10
20
30
40
50
60
70
Time (minutes)
Conc (ug/mL)
Concentration_str
128.0
64.0
32.0
16.0
8.0
4.0
2.0
1.0
0.5
0.25
Row
Row 1
Row 2

Working With Well-plates: Circle Detection

- ROI-based
- Applies Hough Circle Transform locally within the ROI to find circles.



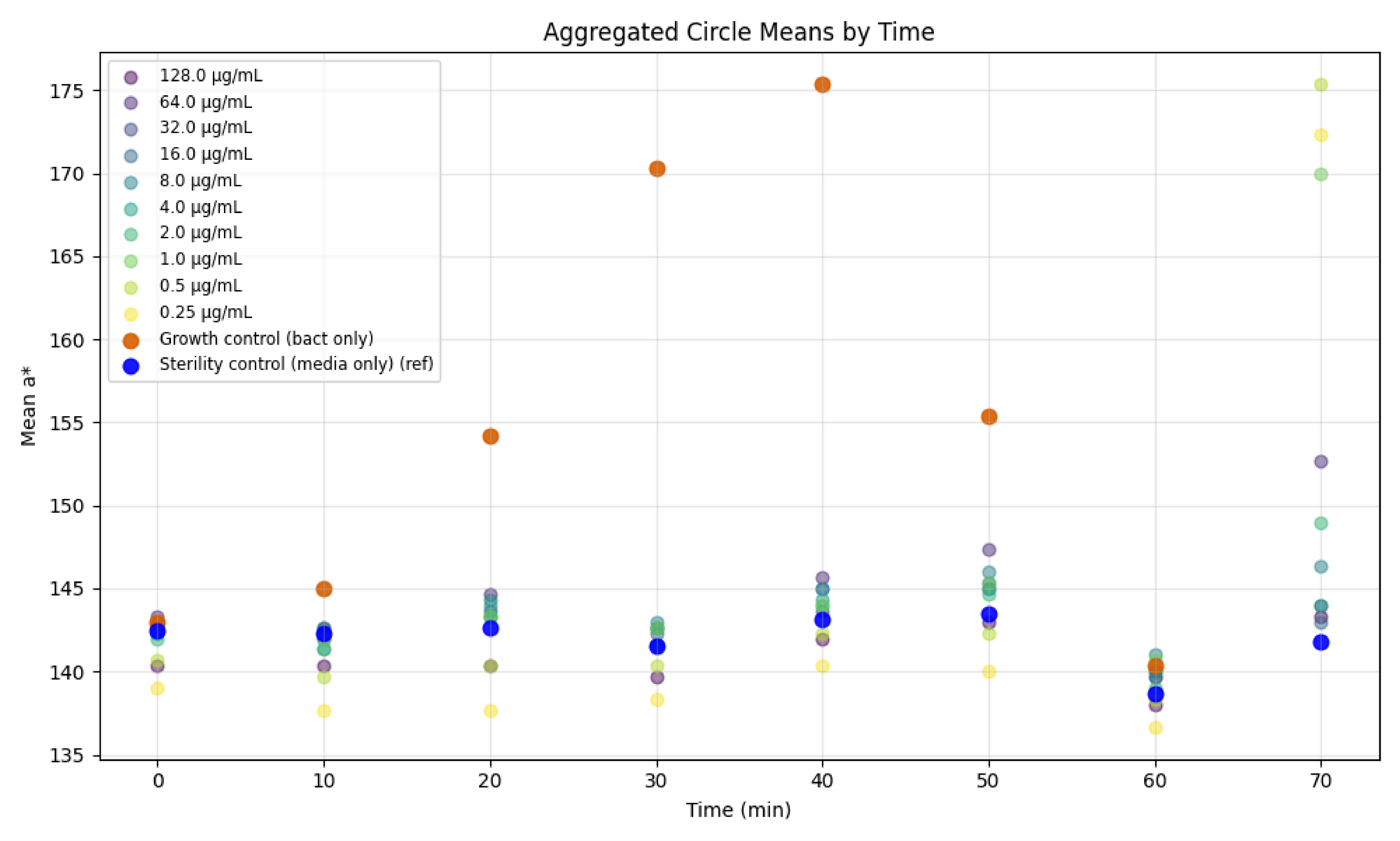


Method: Layout-Based a* Normalization

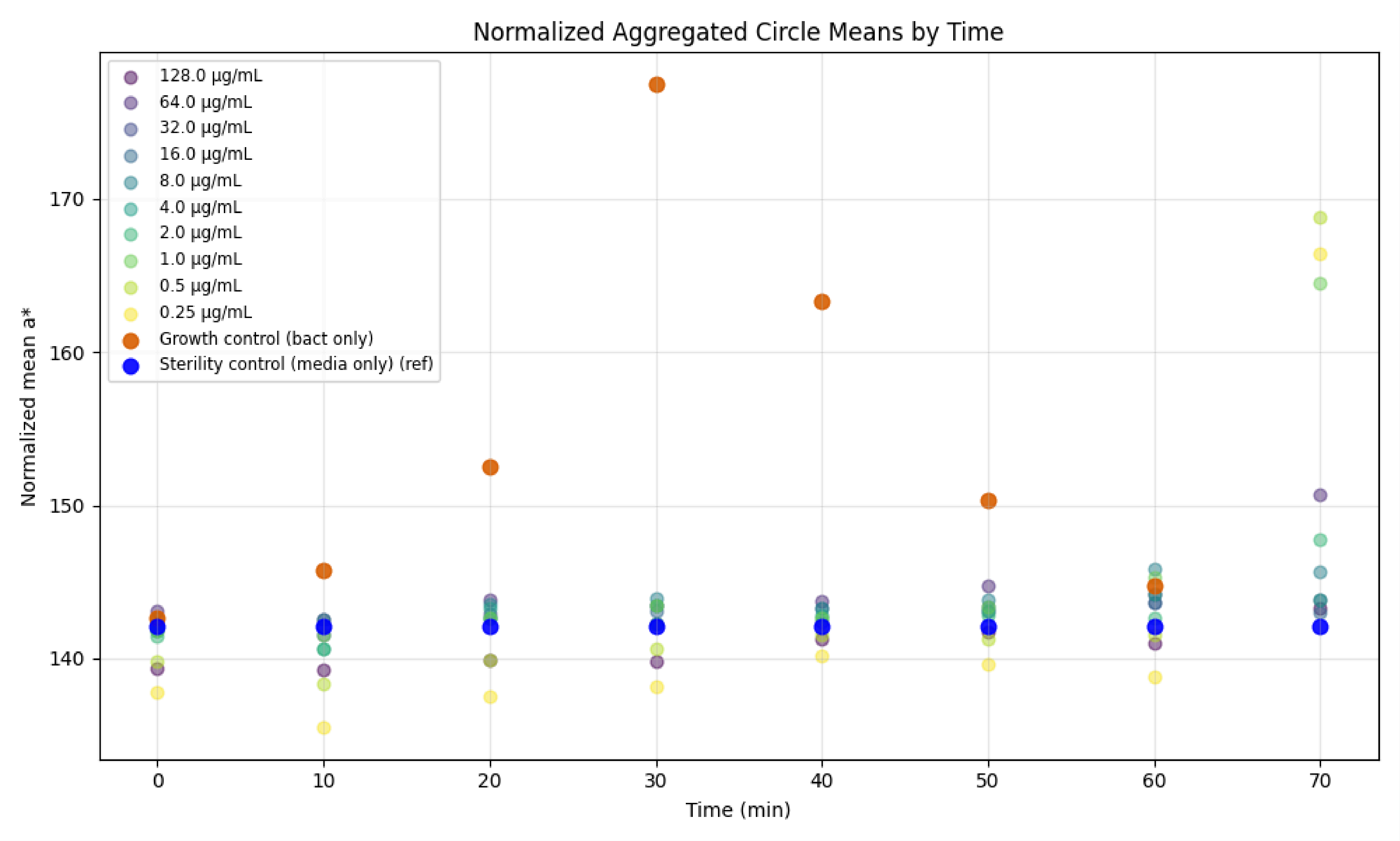
- Normalize well-wise **a*** intensity at each timepoint using a fixed plate layout and the control wells to account for lighting conditions from image to image.

1. Reference extraction from plate layout; it is the lower-right 2x3 wells (media only control)
    - Only fully complete layouts (10/10/10/6/6) contributed to global reference statistics.
    - We computed:
    - Global mean of reference values, **μ_all**
    - Global target spread, **σ_all** = (mean of per-timepoint reference standard deviations)
2. Per-timepoint affine normalization
    - For each timepoint t with reference mean **μ_t** and std **σ_t** (i.e., of the reference samples at timepoint t):
        - Scale: **$s_t = \sigma_{all} / \sigma_t$ (or 1 if $\sigma_t = 0$)**
        - Offset: **$b_t = \mu_{all} - s_t \mu_t$**
        - Normalized value (of every sample at timepoint t): **$a^*_{norm} = s_t \cdot a^* + b_t$**
3. Output
    - Time-series scatter plot of normalized group means (dose columns + controls)

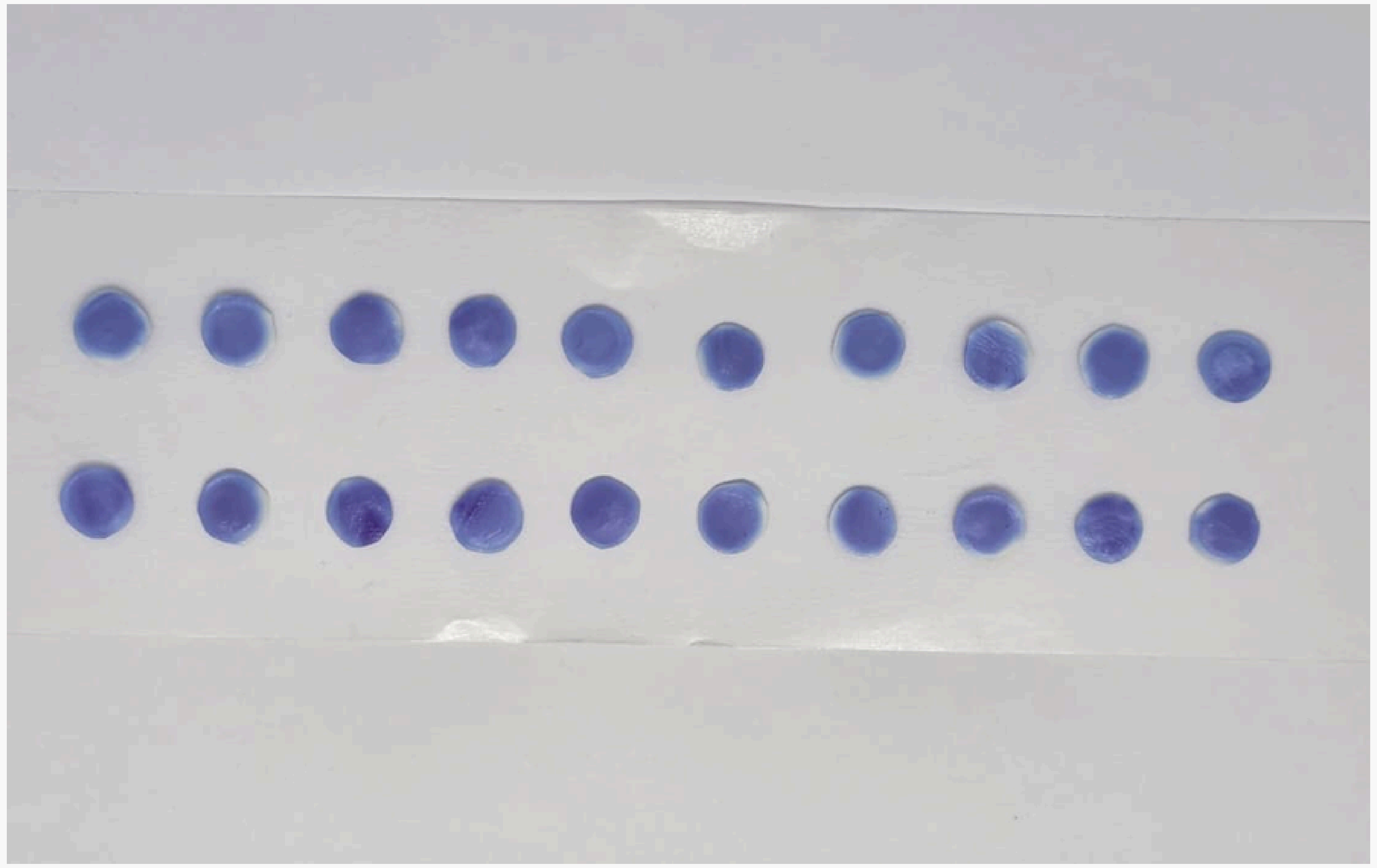
Results (Without Normalization)



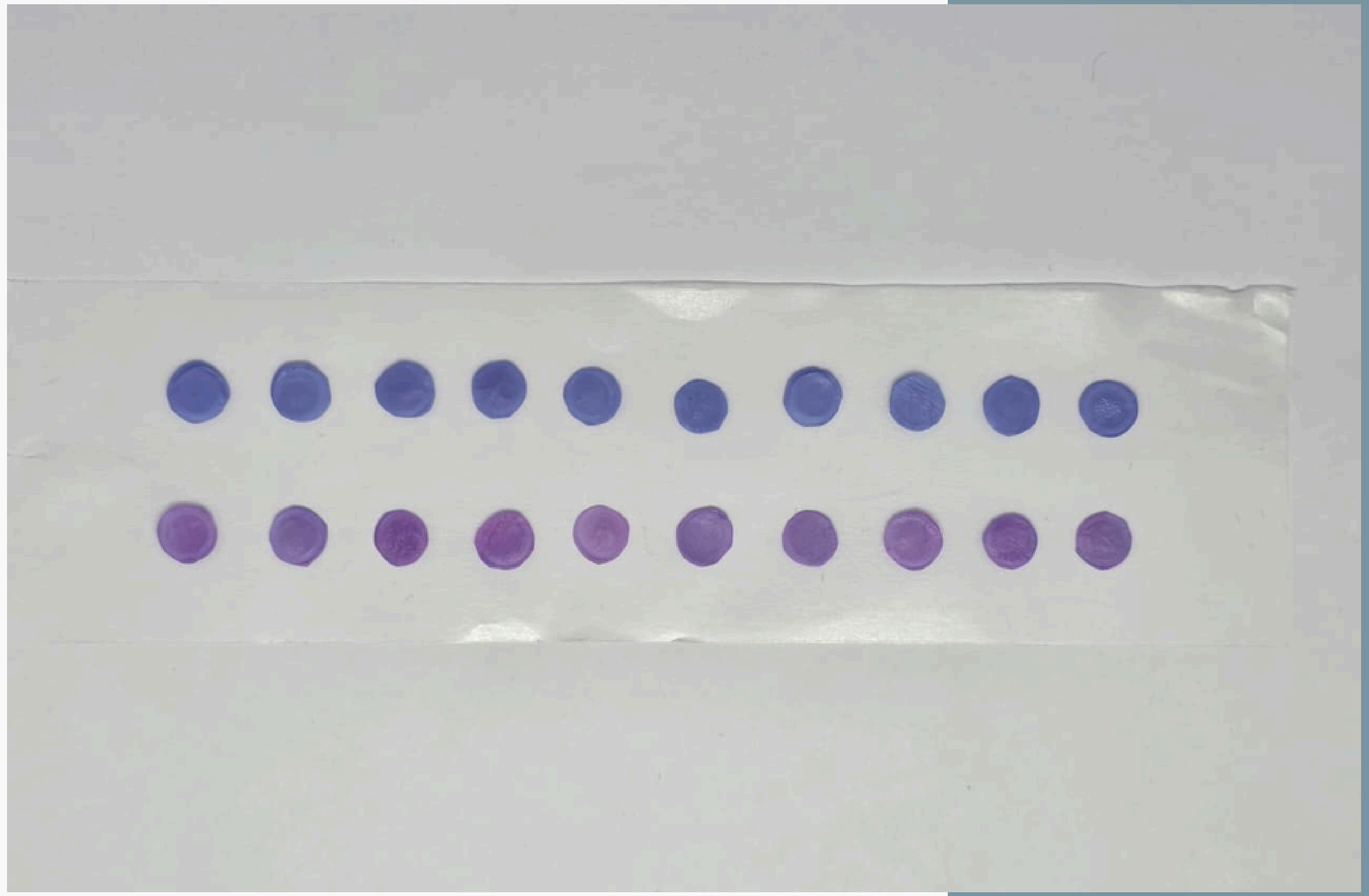
Results (With Normalization)



Freeze-dried Bacterial Cellulose



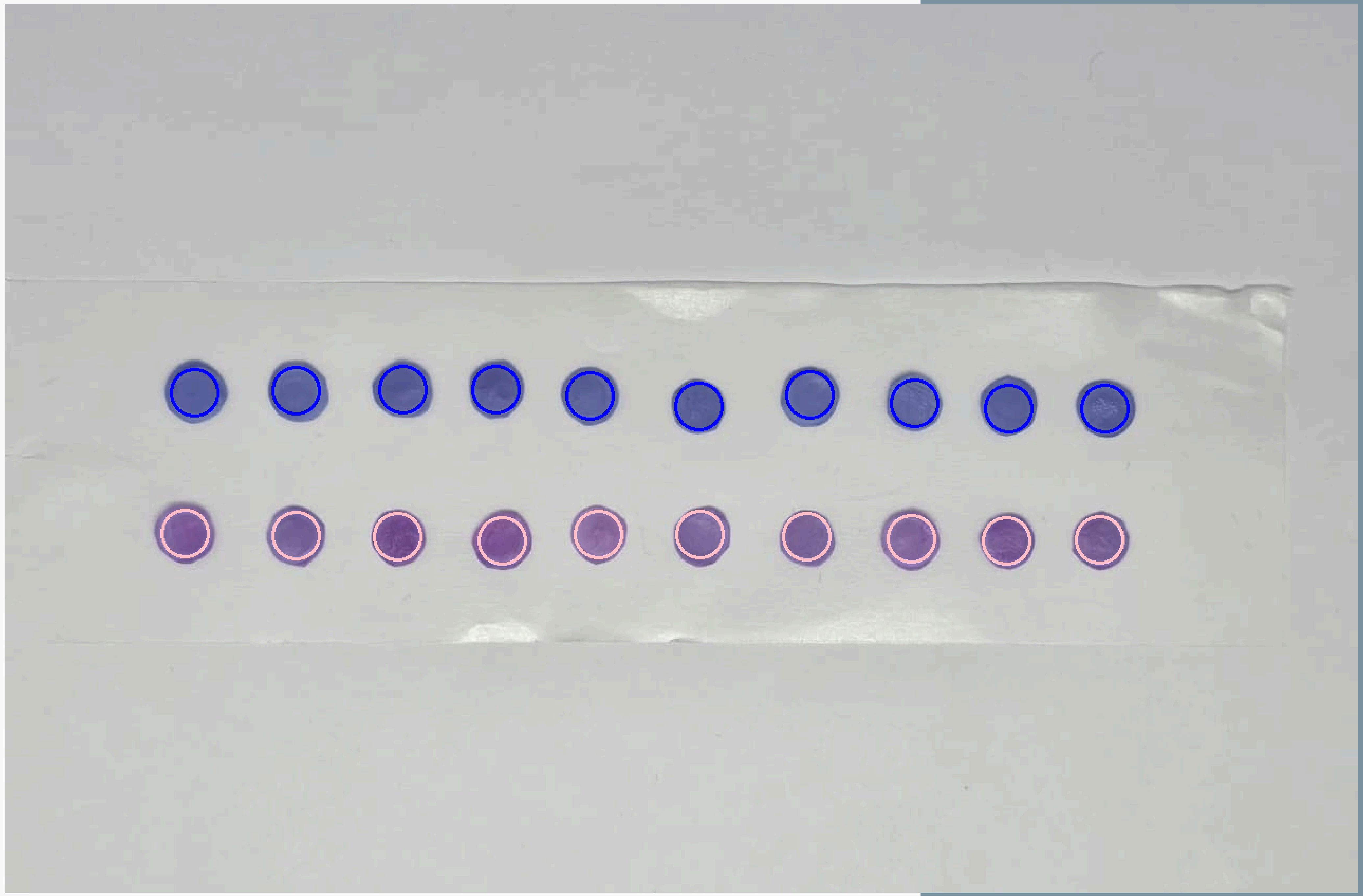
0th min



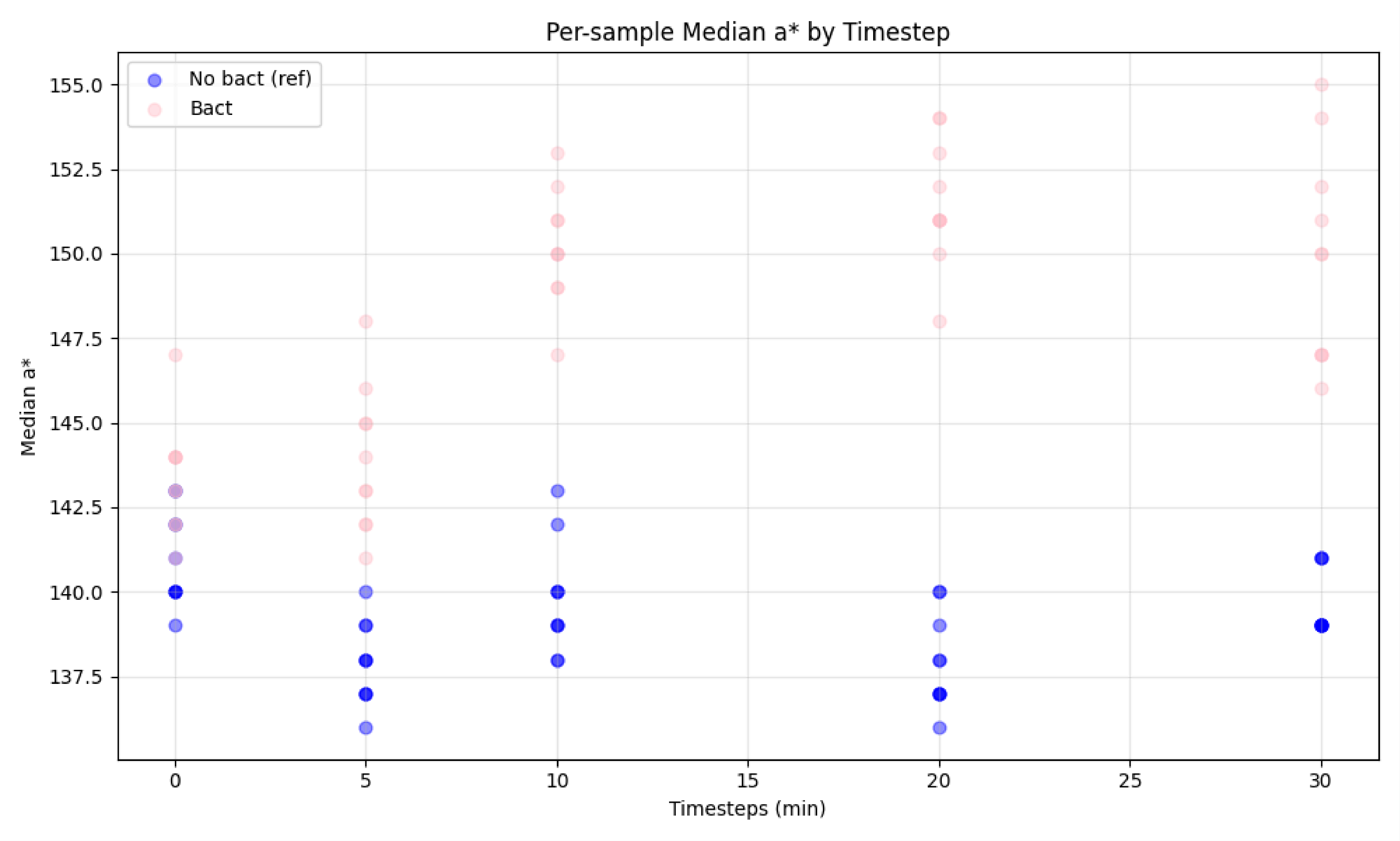
25th min

Working With FDBC

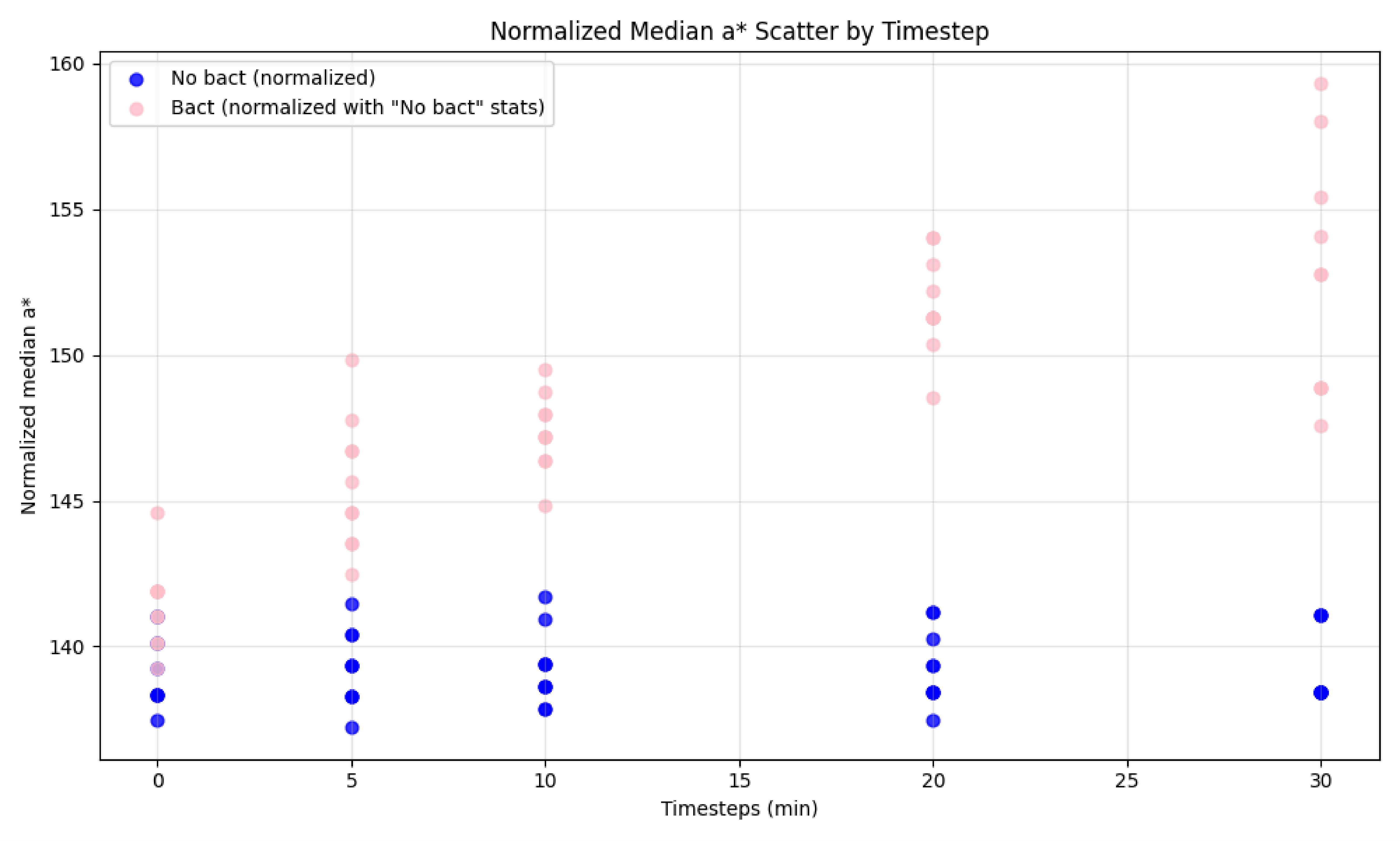
- Circle detection using SAM



Results (Without Normalization)



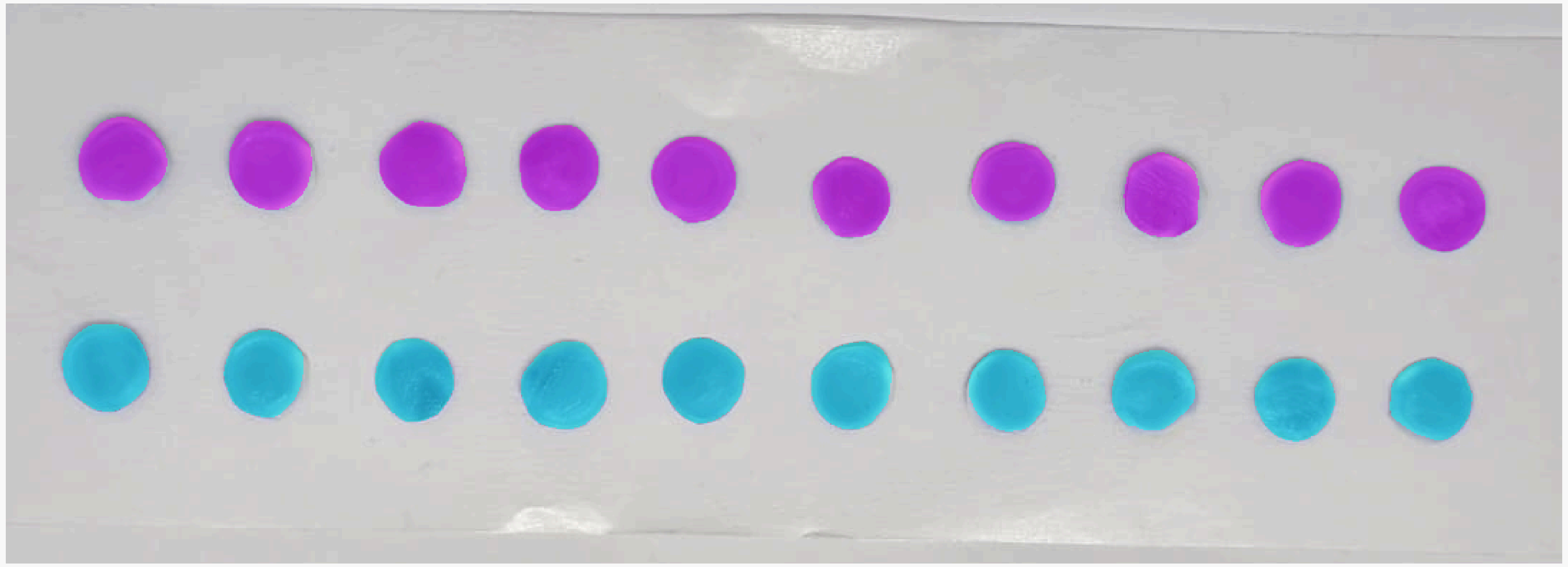
Results (With Normalization)



Coming to FDBC...

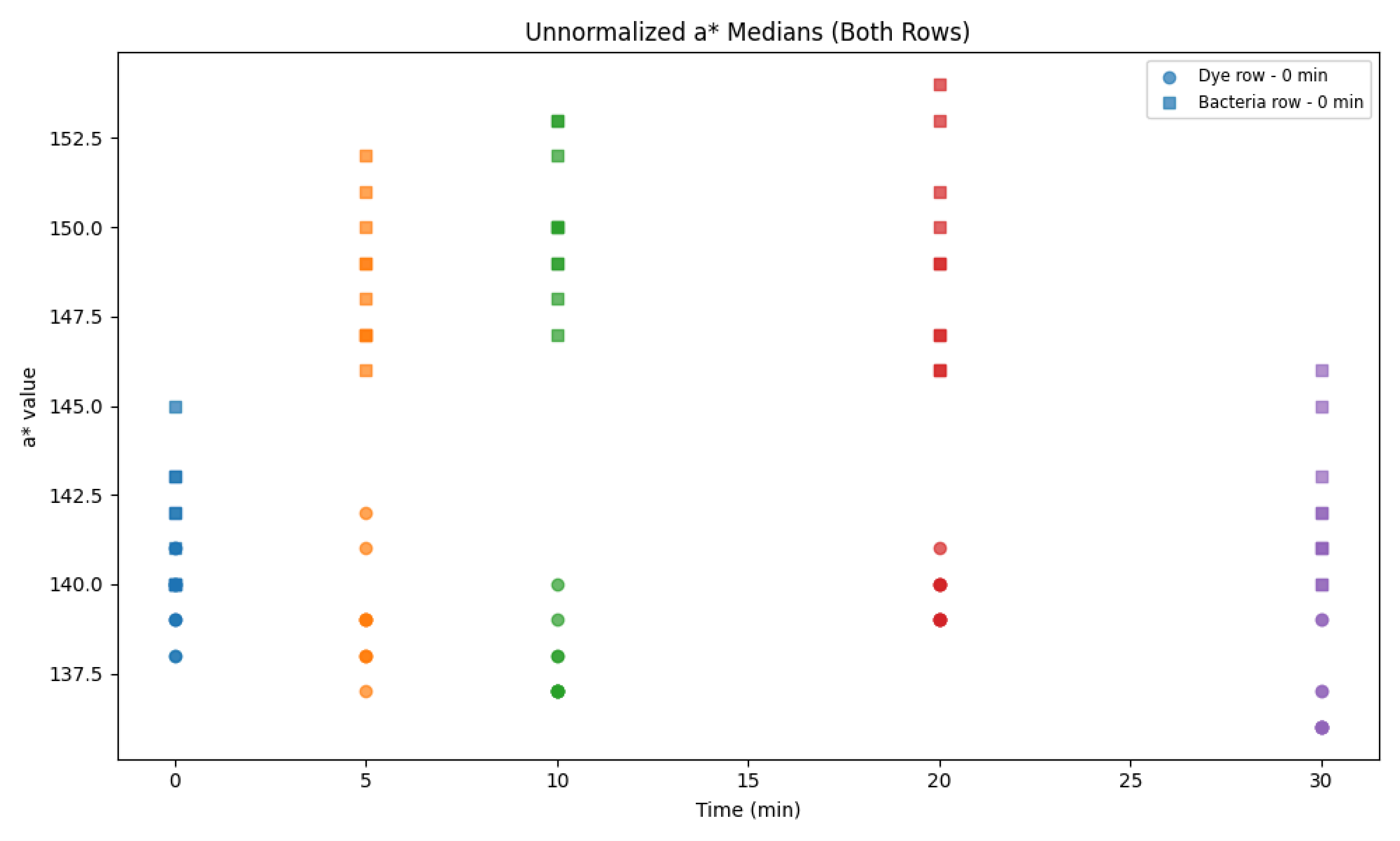
- We have only control samples - a row without bacteria, and a row with.
- The median is chosen as the representative for each disk.
- Scatter plots serve as a basic indicator of how the two rows' pinkness compare to each other, how they change over time, and how much intra-row variance exists.
- We also plot KDE plots, to indicate the general distributions the disks obey.
- The medians serve as the basis for the following:
  - We keep track of various row statistics - mean, IQR, etc.
  - We track the changes of the row distributions using Jensen-Shannon divergence.
  - We perform the KS hypothesis test.

The Current Setup

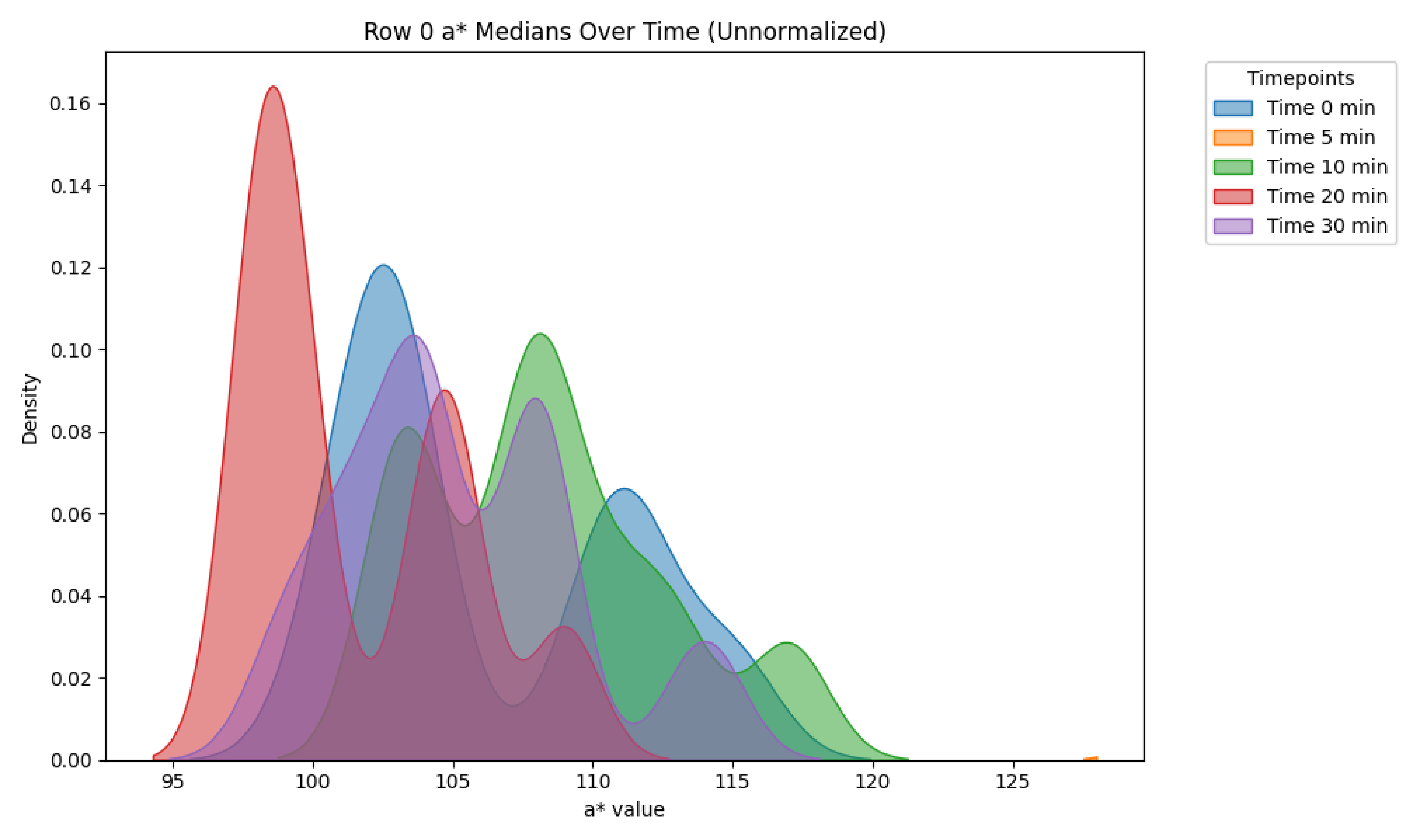


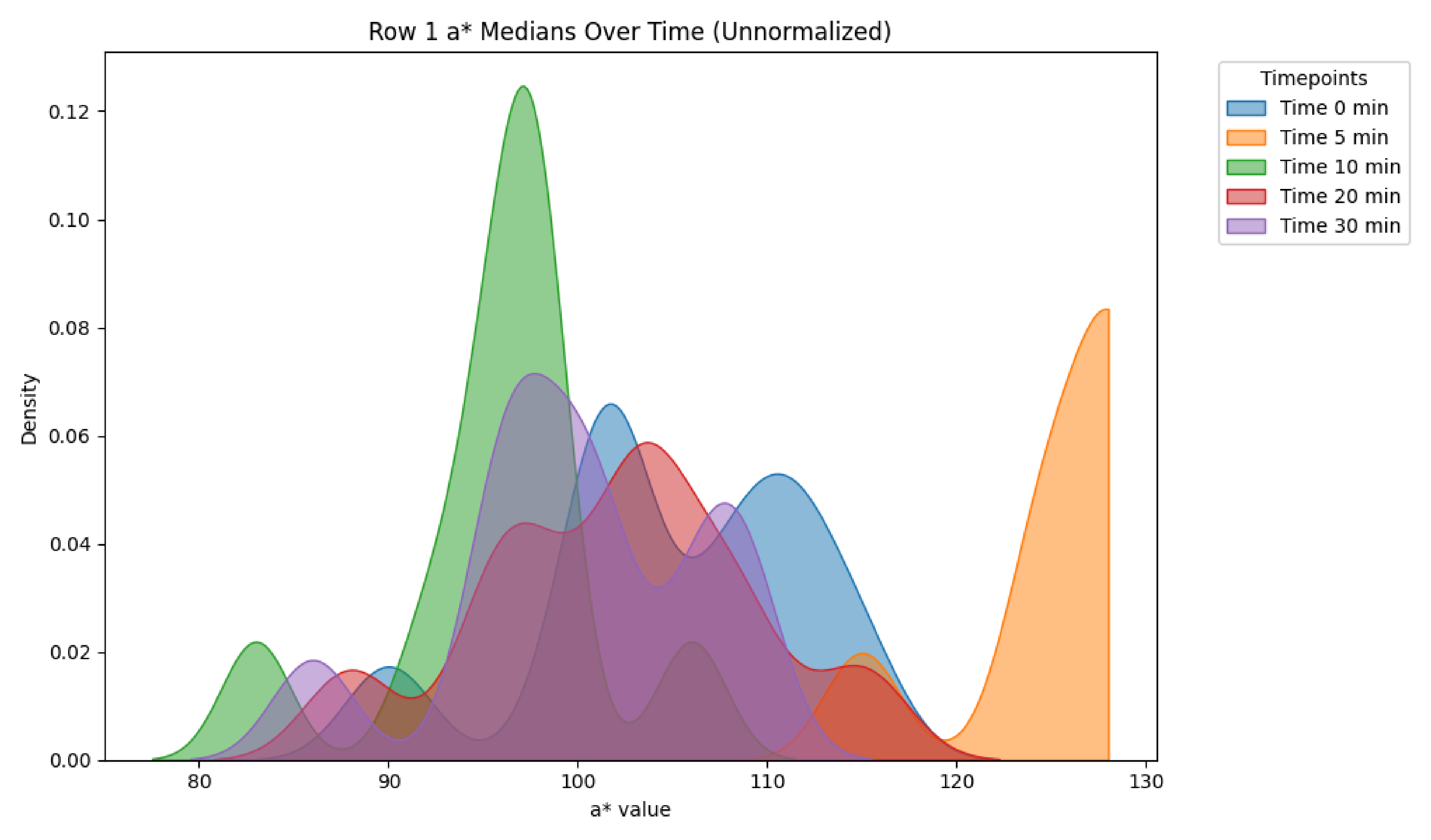
- We take 10 disks per concentration, to support our probabilistic model.
- We detect rough circle centers with the Hough transform, and then use SAM.

Visuals...


Unnormalized a* Medians (Both Rows)
Dye row - 0 min
Bacteria row - 0 min
152.5
150.0
147.5
145.0
142.5
140.0
137.5
a* value
0
5
10
15
20
25
30
Time (min)

More visuals...

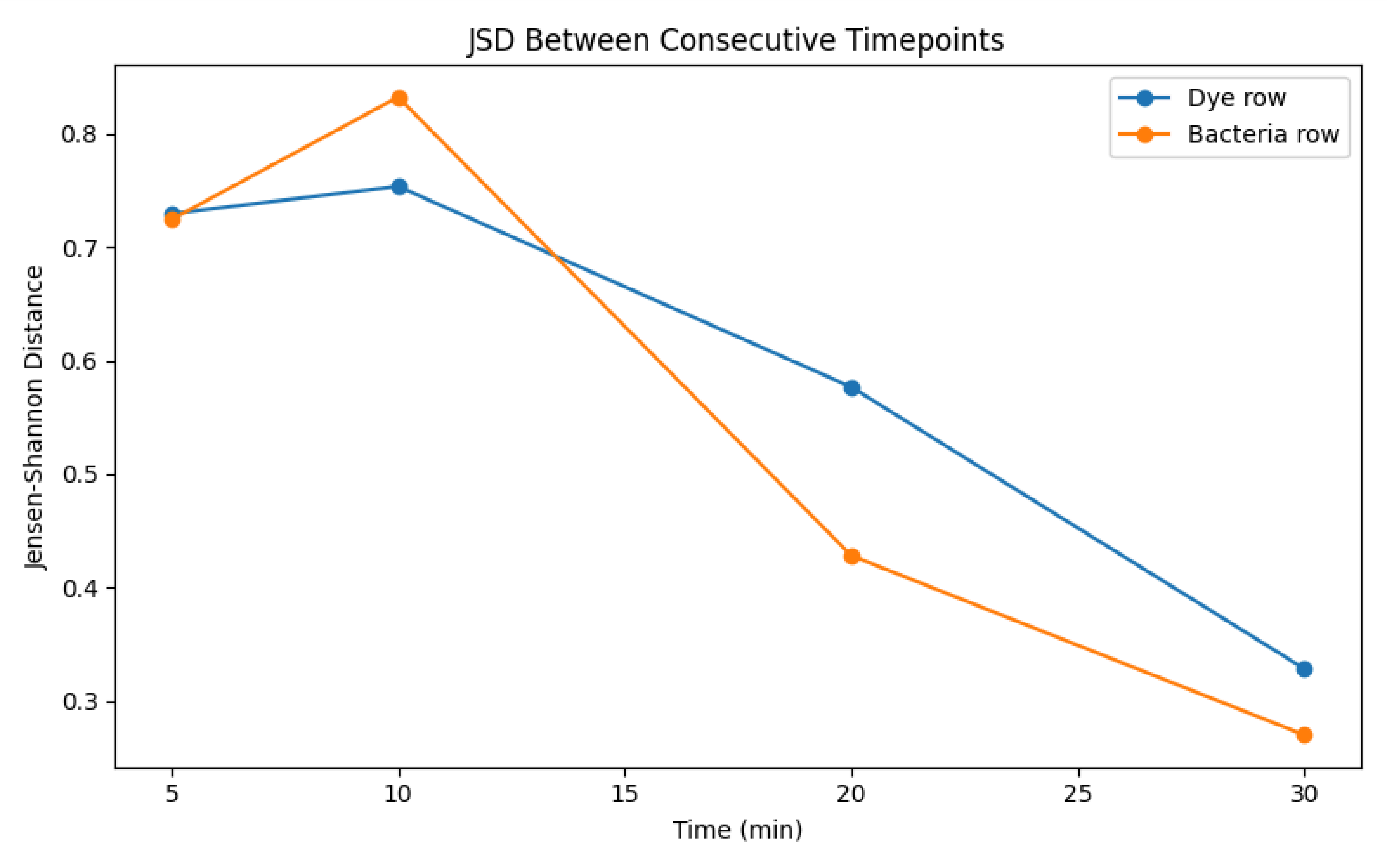




Jensen-Shannon Distance

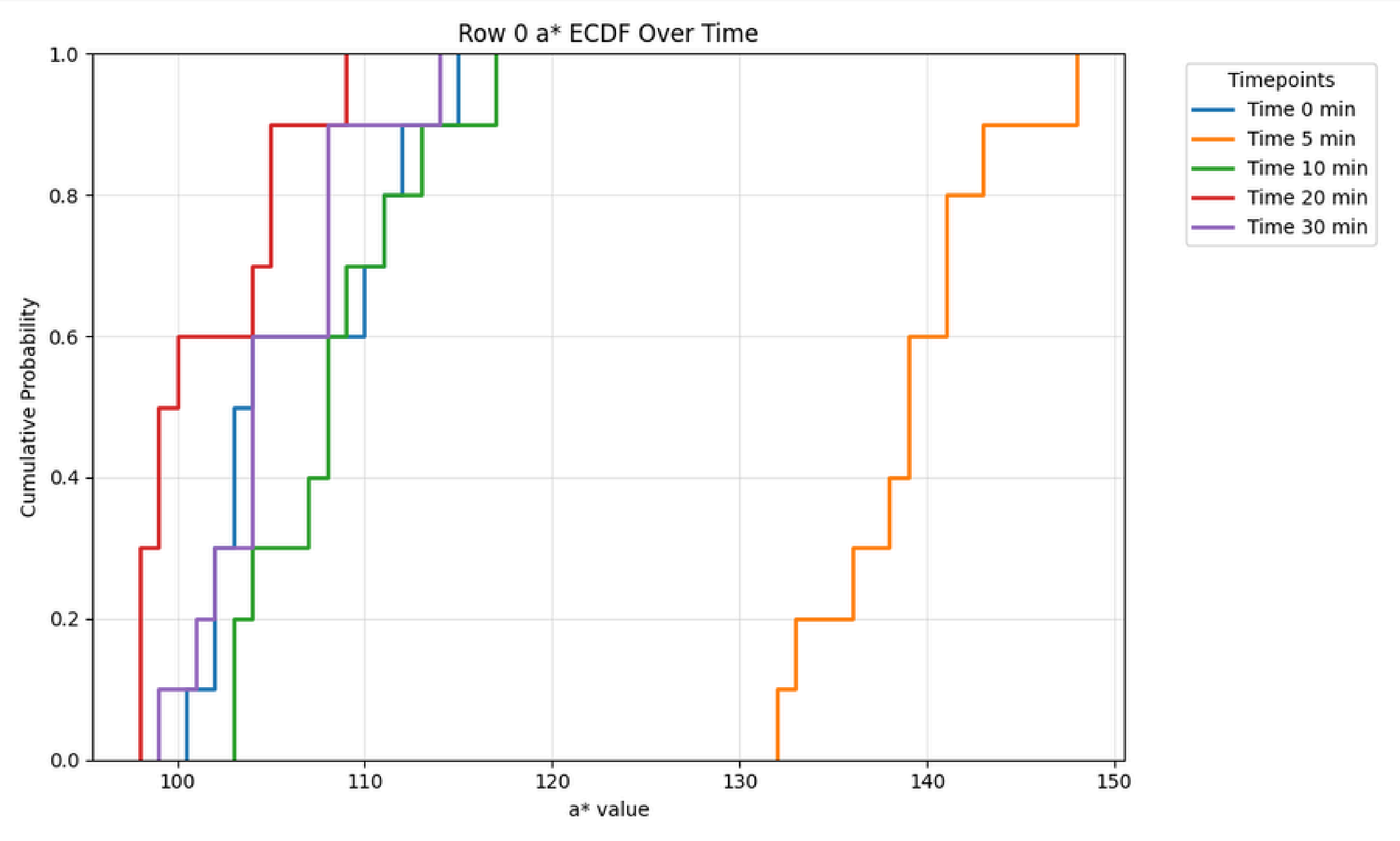
- Definition: A method of measuring the similarity between two probability distributions.
  - It is the square root of the Jensen-Shannon Divergence and is preferred over KL-Divergence because it is symmetric and always yields a finite value.
  - It is the square root of the average of the KL-Divergences between 2 distributions and their mean distribution.

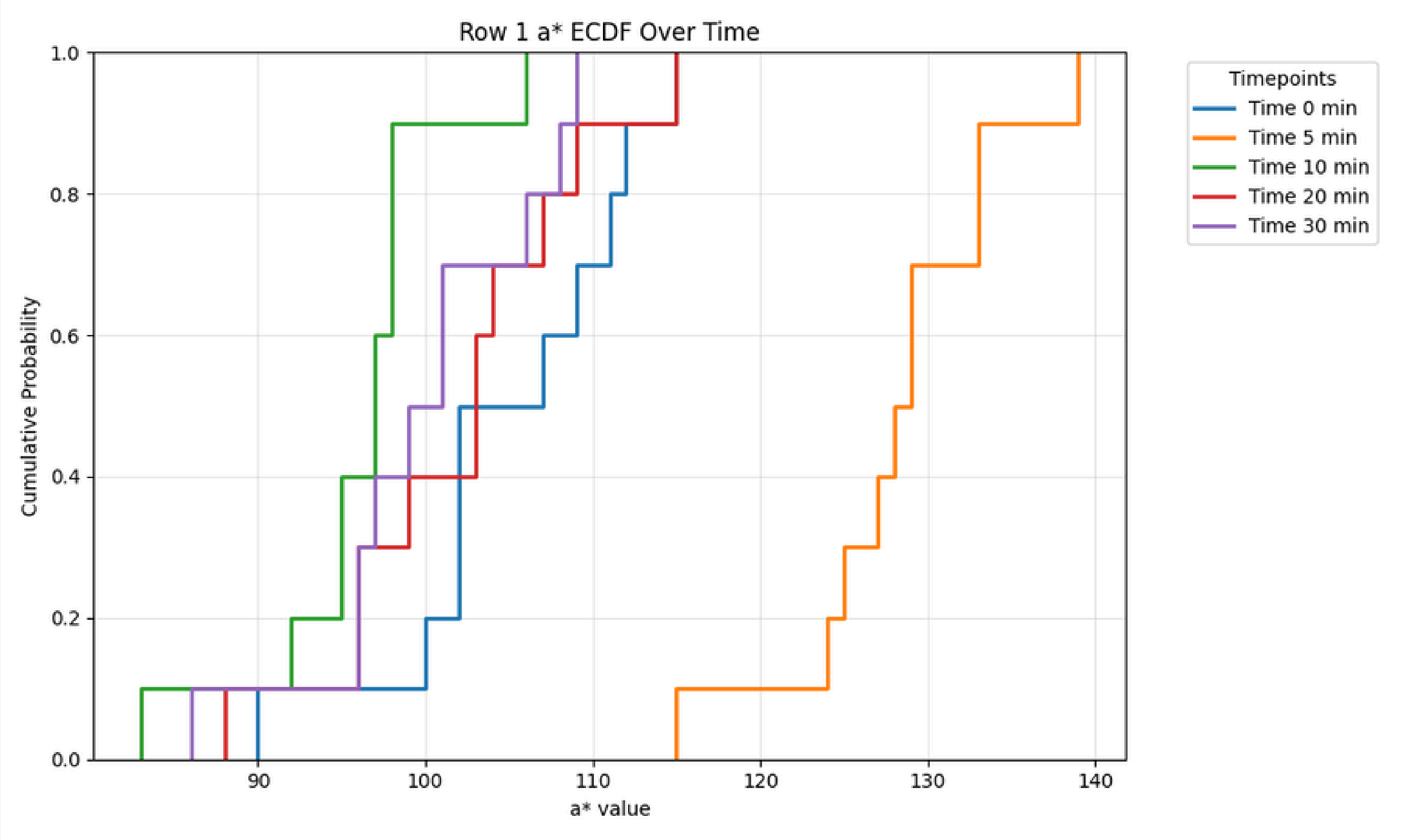
JSD Visual


JSD Between Consecutive Timepoints
Dye row
Bacteria row
Jensen-Shannon Distance
0.8
0.7
0.6
0.5
0.4
0.3
5
10
15
20
25
30
Time (min)

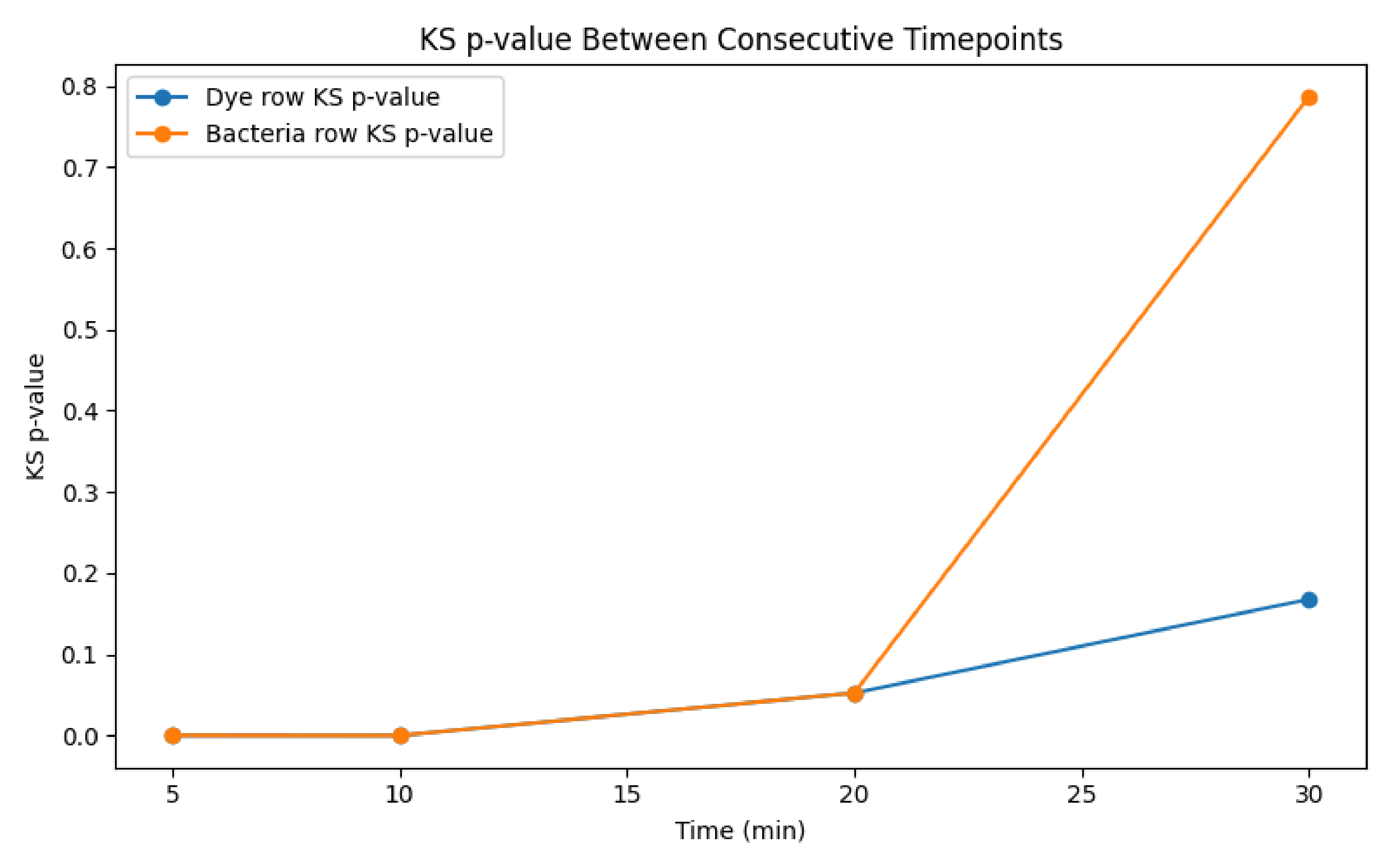
Kolmogorov-Smirnov Test

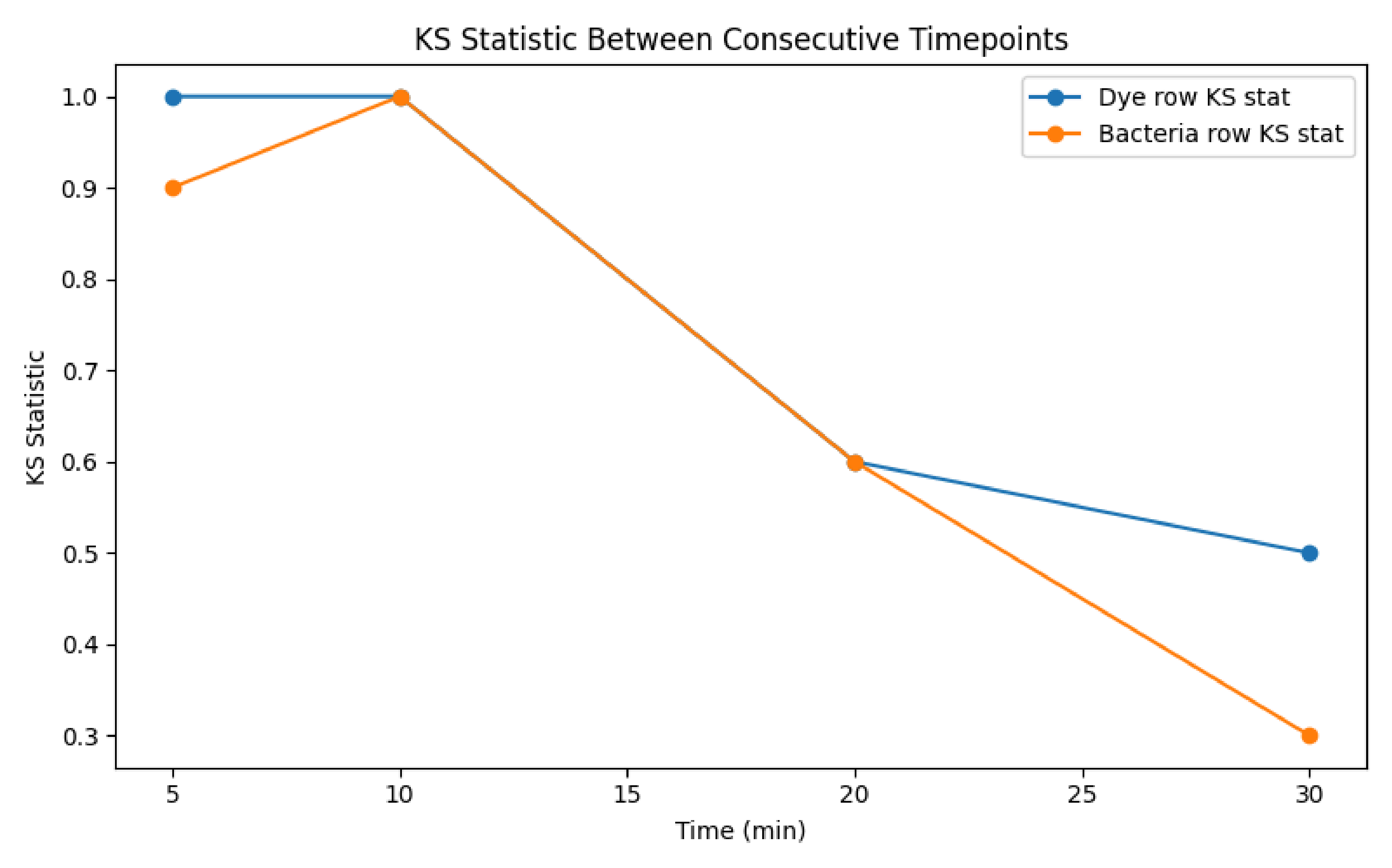
- We calculate the distances between the empirical CDFs of the 2 rows.
- This is another way of tracking the distribution change over time, and is compared and contrasted with the JSD.





Distance & P-Value





Conclusion

- **Extensive Methodological Testing**: We explored and implemented a wide array of computer vision pipelines and statistical tracking methods (Signal-Guided SAM, CIELAB a* tracking, JSD, KS Tests).
- **Promising Initial Signals**: Several of these approaches show strong potential in accurately tracking the colorimetric shift (Resazurin to Resorufin) and isolating relevant data.
- **The Generalization Challenge**: While early indicators are highly positive, we must be cautious of our limited sample size.

Future Work

- We need to get samples for at least 2 concentrations of antibiotic.
- We extend the current testing methodology to see if it holds when the reactions actually take place.
- This could yield information regarding the behavior of the test across different conditions.
- We hope to obtain enough information to identify whether the test has failed/become defective early on.
    - Conversely, predicting success early on is also a target.
    - We also hope to predict whether the concentration of antibiotic is below/above MIC by tracking the reaction progress, providing an early result.

Thank you